KrF_IB models: accuracy, energy ladders and reaction rates

LIGHT v0_5 (Kr/F_2, the KrF laser gas) and LIGHT_Kr v0_5 (pure krypton), in `code/LIGHT/v0_5/` and `code/Kr_LIGHT/v0_5/`, 20 October 2026

Bottom line.

- **LIGHT v0_5** (59 rows on 12 species). It reproduces the KrF(B,C) decays its rate fit used within 6.1 % rms, and the decays no fit used within 11.7 % rms. At the Tahoma laser it gives the small-signal gain to −6 %, the absorption to −25 %, the net gain to −2 %, the saturation intensity to +4 % and the formation efficiency to −3 %. **The propagated uncertainty of its own inputs is ±40 %**: ±40 % on Tahoma's net gain, ±38 % on the 248-nm share (5.68 % of E_{dep}, ±2.17 points), ±33 and ±37 % on Tahoma's gain and absorption. Its largest errors:
  - Hakuta et al.'s C–A decays, 19.3 and 34.1 % rms;
  - Burnham and Searles' KrF(B) lifetime, −22.3 %;
  - Kleikamp et al.'s Kr pressure optimum, −37 %, and their F_2 optimum not reproduced at all.
- **LIGHT_Kr v0_5** (219 rows on 10 species). The parts that measurements reach are close:
  - drift velocity within 2.8 % rms;
  - ionization coefficient within 9.9 % rms at 50–300 Td;
  - electroluminescence yield at 0.99–1.04 of the measurement;
  - e-beam VUV efficiency +29 % (low pump) and −3 % (high pump).

  Its headline output is not close. The 147-nm share at the reference point is 6.08 % of E_{dep} with **a propagated uncertainty of ±115 %** (±7.0 points). That term is dominated by electron collisions with Kr_2^*, which nothing measures, and no measurement tests the field-driven regime the model is built for.

Both models are 0-D and predictive only: no rate is tuned to a device measurement. Device and beam data only score the model; LIGHT's KrF(B,C) constants are a joint fit to decay data, and LIGHT_Kr's cross sections are trained on swarm data alone. The energy ladders are in Sections 1.4 and 2.4, and every reaction rate is in Appendices A and B.

	LIGHT v0_5	LIGHT_Kr v0_5
gas	Kr/F_2 (0.5 % F_2 at the reference point)	pure Kr
code	`code/LIGHT/v0_5/` (`run_v0_5.py`)	`code/Kr_LIGHT/v0_5/` (`run_v0_5.py`)
network	59 rows, 12 species, 17 bands (11 rows at 0 by default)	219 rows, 10 species, 22 bands (31 rows at 0 by default)
electrons	kinetic EEDF (two-term Boltzmann with e–e collisions) by default; two-temperature closure as the fast option	the same
reference point	1 atm, 300 kW cm^{-3} for 1 µs, run to 2 µs	the same, pure Kr
headline output	248 nm: 5.73 % of E_{dep} (kinetic), 5.7 % (two-temp.)	147 nm: 6.08 % of E_{dep} (kinetic), 5.58 % (two-temp.)
its uncertainty	**±40 %** (quadrature, 22 inputs; ±38 % on the 248-nm share)	**±115 %** (quadrature, 24 inputs and the cross sections' posterior)
what v0_5 adds	K76 ($e^- + Kr^* \rightarrow e^- + Kr^{**}$) on Jung et al.'s measured cross sections; Kleikamp et al.'s laser held out	the electrons' collisions with excited Kr and Kr_2^* (Vriens–Smeets), each 5s→5p pair at its own threshold
accuracy source	`LIGHT/docs/accuracy_overview_v0_5.pdf`	`Kr_LIGHT/docs/accuracy_overview_v0_5.pdf`

Table 1: The two models at a glance.

1 LIGHT v0_5: the Kr/F_2 model

What it is. LIGHT is a 0-D model of KrF laser gas. It is one uniform box at constant volume, and a given power goes into its electrons. It follows that energy into excitation, ionization, KrF(B,C) and Kr_2F^* formation, the 248, 272, 275 and 420-nm and 147-nm bands, heat and the electron gas, and its energy ledger closes by construction. v0_5 is v0_4 with K76 ($e^- + Kr^* \rightarrow e^- + Kr^{**}$) on the cross sections Jung et al. measured out of the metastable levels, in place of a 1976 fit, and with a second laser, Kleikamp et al.'s, held out. Everything here is from v0_5's generated accuracy overview.

1.1 Against the data

quantity	measured	LIGHT	error	role
energy ledger closure		$< 3.4 \times 10^{-14}$		bookkeeping
KrF(B) decay vs Kr and F_2, Eden 1978			rms 2.3, 3.2 %	trains
KrF(B,C) slow decay, Gadomski 1992 Figs. 1, 2			rms 6.1, 4.2 %	trains
C/B emission ratio at 300 K, Yu & Setser 1990	0.16 ± 0.02		-0.5 %	trains
KrF(B,C) decays, Xu 1993 Figs. 5–7	three figures		rms 11.7 %	held out
decay constants at 50, 100 Torr Kr, Xu 1993	5.0, 6.2×10^7 s^{-1}	$\times 0.91$, $\times 0.97$	-9, -3 %	structure check
C/B ratio 230 K over 300 K, Yu & Setser	1.30 ± 0.06		-8.4 %	held out
C–A decays, Hakuta 1990	two figures		rms 19.3, 34.1 %	held out, in conflict
τ_B, Burnham & Searles 1977	9.0 ± 0.6 ns		-22.3 %	held out, in conflict
Tahoma small-signal gain g_0, kinetic EEDF	6.39 %/cm	5.99	-6 %	device, score only
Tahoma absorption α_0	1.11 %/cm	0.83	-25 %	device, score only
Tahoma net gain $g_0 - \alpha_0$	5.28 %/cm	5.16	-2 %	device, score only
Tahoma saturation intensity (two-temp.)	1.4 ± 0.1 MW/cm^2	1.45	+4 %	device, score only
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$	0.27	0.261	-3 %	device, score only
Tahoma loss in pure Kr, 257.3 nm	0.67 %/cm	0.24	$\times 0.36$	rough check
VUV yield of pure Kr at low pump, Morozov 2008	0.42 ± 0.05	0.55	+32 %	held out (v0_3's run)
Kr pressure optimum, Kleikamp 1986, F_2 10 mbar	2 atm	1.26 atm	-37 %	device, held out
F_2 pressure optimum, Kr 2 atm	10 mbar	> 30 mbar	not reproduced	device, held out
three-body KrF^* + Kr quenching their balance deduces	4.6×10^{-31} cm^6/s	1.02×10^{-30}	$\times 2.2$	device, score only

Table 2: LIGHT v0_5 against the data, every band at its default. An error is model less measurement over the measurement, signed for one value and rms over a set. "Trains": the KrF(B,C) constants are a joint fit to these decays; every device row only scores the model. Tahoma's saturation intensity and formation efficiency and the Kleikamp optimum run on the two-temperature closure; the kinetic EEDF moves Tahoma's numbers by about 1 %.

1.2 Propagated uncertainty: ±40 %

Each of 22 inputs is set alone to the two ends its source allows: 17 bands, 3 measured constants with quoted errors and 2 cross sections. Half the difference between the two runs is that input's term, and the terms add in quadrature. That sum is indicative, not a confidence interval, and the summed excursions are the ceiling of the spread.

output	value	quadrature	largest terms
248-nm share	5.68 % of E_{dep}	±2.17 pts (±38 %)	the trimer's $(T_g/300)^n$ (±1.70), Kr_2^* + F → KrF(B) (±0.91), KrF(B,C) + F (±0.88), e^- + Kr* ionization (±0.29); excursions −6.85/+2.46 pts
Tahoma g_0	5.93 %/cm	±33 %	trimer law (±1.55), KrF* per ion-ion recombination (±0.75), σ_{se} (±0.71), KrF(B,C) + F (±0.47); measurement 0.2 units away
Tahoma α_0	0.82 %/cm	±37 %	Kr_2F^* + F_2 (±0.18), Kr_2F^* + F (±0.16), $\sigma(Kr_2F^*)$ (±0.12), trimer law (±0.10); measurement 0.9 units away
Tahoma $g_0 - \alpha_0$	5.11 %/cm	**±40 %**	measurement 0.1 units away

Table 3: LIGHT's propagated uncertainty (two-temperature closure). Every Tahoma measurement lies within one quadrature unit. Not propagated: the electron-impact cross sections (the band k_5s5p carries the 5s→5p set's ±35–40 %) and the fitted KrF(B,C) set, which the training decays pin at 6.1 % rms.

1.3 The reference point

At 1 atm, 0.5 % F_2, 300 kW cm^{-3} for 1 µs, run to 2 µs:

percent of E_{dep} unless stated	kinetic EEDF	two-temperature
248 nm, KrF B–X	5.73	5.7
272 nm, KrF B–A / 275 nm, C–A	0.183 / 0.829	0.2 / 0.8
420 nm, Kr_2F^*	4.21	3.9
147 nm, Kr_2^*	4.8	4.6
near IR, 5p→5s	0.28	0.3
heat, reactions / elastic	72.6 / 1.8	71.6 / 3.4
KrF(B) formed	24.2	23.0
g_0 / α, 500–1000 ns (%/cm)	3.79 / 0.67	3.96 / 0.64
T_g at the end (K)	738	742
T_e, pulse mean (eV)	2.96	5.68

Table 4: LIGHT's reference point. What moves it is the 5s↔5p cycle: the electrons pump 79–81 % of the Kr* formed up to the 5p rung, which relaxes on Kr, and each cycle's 1.385 eV heats the gas (33.7 % of the input). Numerically, the ledger closes below 10^{-14} and the kinetic and model energy balances agree to 6.6×10^{-6}.

1.4 The energy ladder

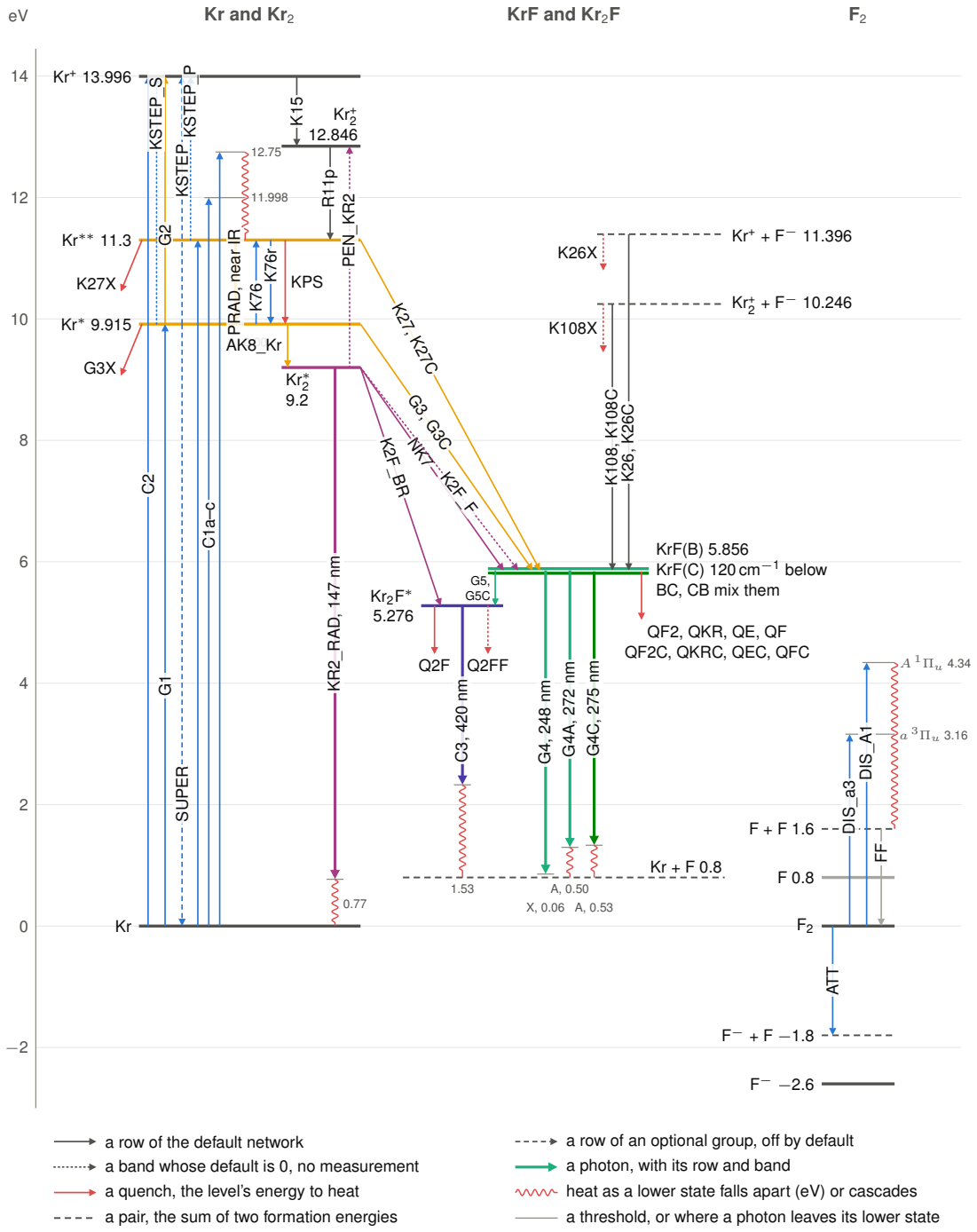


Figure 1: LIGHT v0_5's energy ladder, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair; an arrow is a row, labelled with its id (Appendix A). Solid: default network; dotted: a band whose default is 0 (Q2FF, K26C, K26X, K108C, K108X, K2F_F, PEN_KR2, KSTEP_S, KSTEP_P, QF, QFC); dashed: the optional group `electron`. A photon's arrow drops by its quantum and the red squiggle is the heat its lower state gives the gas; a red stub is a quench. KrF(B) lies 5.54 eV below $Kr^+ + F^-$ and 5.06 eV above Kr + F, and the 248-nm photon carries 5.00 eV of that. The 5p rung lies 1.39 eV above Kr^*. G6 and VIB1–4 move nothing between levels and are not drawn. From `LIGHT/docs/accuracy_overview_v0_5.pdf`, Figure 1.

species	E_{form} (eV)
Kr^{+}	13.996
Kr_2^{+}	12.846
Kr^{**}	11.300
Kr^{*}	9.915
Kr_2^{*}	9.200
KrF(B)	5.856
KrF(C)	5.841
Kr_2F^{*}	5.276
F	0.800
Kr	0.000
F_2	0.000
F^{-}	-2.600

Table 5: The rungs of LIGHT's ladder as the code books them (`species.e_form`): formation energy above ground-state Kr and F_2, which every row's energy release is the difference of.

1.5 Where LIGHT is not yet reliable

- The trimer's rate at laser gas temperatures. Its temperature law, a calculation measured only to 425 K, is the largest term of the propagated uncertainty. Kleikamp et al.'s laser finds LIGHT's quenching $\times 2.2$ too strong at 362 K, through a model-dependent constant.
- The F_2-side absorbers and KrF^{*} + F_2. The laser's F_2 optimum is not reproduced.
- The rows with no measurement. Kr_2^{*} + F, KrF(B,C) + F, Kr_2F^{*} + F, Kr_2^{*} + Kr_2^{*} and the stepwise ionizations are bands at 0.
- Two J = 1 5s levels have no measured 5s→5p cross section and use proxies; Jung et al.'s absolute scale is ±35–40 %.
- The steady EEDF after the pulse, Hakuta's and Burnham & Searles' conflicts, the absorbers as a diagnostic only, no cavity, and 0-D.

2 LIGHT_Kr v0_5: the pure-krypton model

What it is. LIGHT_Kr is LIGHT with every fluorine species and row removed, rebuilt for pure krypton. Its network has these parts:

- all 51 of Biagi's excitation levels at their own thresholds, with cross sections trained on swarm data;
- the four 5s levels as separate species;
- every measured 5s→5p pair at its own threshold;
- excimer formation out of each 5s level, and its redissociation by detailed balance;
- Signorell's Kr_2^{+}.

v0_5 adds the electrons' collisions with the excited atoms and with Kr_2^{*}. Where nothing is measured, the ionizations use Vriens and Smeets's calculation, as bands from 0 to twice it. Kr_2^{*}'s quenching and state mixing by electrons are bands at 0. That cut the reference point's 147-nm share from v0_4's 31.22 % to 6.08 %.

2.1 Against the data

test	measured	error	role
energy ledger closure, reference point		4×10^{-15} of E_{dep}	bookkeeping
energy balance at each of 30 nodes		to 10^{-8}	bookkeeping
drift velocity, 3 sets that agree, 79 pts		rms 2.8 %	trains
ionization coefficient 50–300 Td, 5 sets, 64 pts		rms 9.9 %	trains
drift velocity, 8 sets that do not agree (1953–1992)		rms 8 to 47 % by set	held out
characteristic energy D_T/μ, Koizumi 1986, 26 pts		rms 7.1 % (Biagi's file: 15.8 %)	validation
ionization coefficient $<$50 Td, 5 sets, 36 pts		mean -36 % (sets part by $\times 4.0$)	held out by rule
electroluminescence yield 6–13.5 Td, Mano 2022	$0.113\,E/N - 0.3$	0.99 to 1.04 of it	held out
superelastic return of $1s_5$, $1s_4$, $1s_3$, Blagoev 1984	1.5, 5.2, 9.8×10^{-10} cm^3/s	$\times$5.3–6.6, $\times$5.5–9.5, $\times$0.5–1.2	new, held out
Monte Carlo ionization coefficient, same cross sections, 20–50 Td		+3.7 to -4.0 %	solver check
energy per ion pair W under a beam, Saito 2003	23.9 eV	+10 %	held out
excitations per ion pair, Saito 2003	0.55	+24 %	held out
energy per excitation or ion, Saito 2003	15.4 eV	+1.0 %	beam check
VUV efficiency, continuous e-beam, Morozov 2008	0.42 ± 0.05	+29 % (2.5 σ)	held out
VUV efficiency at 3×10^5 W cm^{-3}, Eckstrom 1988	near 0.46	-3 % at 1 atm	held out

Table 6: LIGHT_Kr v0_5 against the data, every band at its default, on the kinetic EEDF. Only swarm data train anything (seven factors on Biagi's cross sections); every other row only scores the model. The swarm rows compare the model as each experiment measures it (two-term solution $\times$ Monte Carlo ratios). No pure-krypton measurement in hand tests the field-driven regime at the reference point's density.

2.2 Propagated uncertainty: $\pm$115 %, and numerical accuracy

Each of 24 inputs is set alone to its two ends, on both closures: 21 bands and 3 measured constants. A calculated row's band runs from 0 to twice the calculation. The trained cross sections add the spread of 12 (kinetic) and 40 (two-temperature) draws of their posterior.

input	its two ends	147 nm, kinetic (% of E_{dep})
$e^- + Kr_2^* \rightarrow Kr_2^+ + 2e^-$, scale on Vriens–Smeets	0 to 2	15.43 to 4.47
$e^- + Kr_2^* \rightarrow e^- + Kr_2^*(0_u^+)$, state mixing	0 to 10^{-7} cm^3/s	6.08 to 12.44
$e^- + Kr_2^* \rightarrow e^- + 2Kr$, quenching	0 to 10^{-7} cm^3/s	6.08 to 1.27
$Kr^*(1s_4) + 2Kr \rightarrow Kr_2^* + Kr$	1.3 to 9.2 $\times 10^{-32}$ cm^6/s	4.29 to 6.08
$e^- + Kr^* \rightarrow Kr^{**}$, scale on Jung et al.	0.6 to 1.4	7.06 to 5.46
$e^- + Kr^* \rightarrow Kr^+ + 2e^-$, scale	0 to 2	6.96 to 5.60
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	3.0 to 17 $\times 10^{-11}$ cm^3/s	7.11 to 6.08
the trained cross sections	12 posterior draws	6.11 $\pm$ 0.06
quadrature sum (24 inputs)		$\pm$7.0 pts (**$\pm$115 %**); two-temp. $\pm$7.2 ($\pm$129 %)
excursions summed by sign		$-$13.2 / +21.5 pts

Table 7: The largest terms of LIGHT_Kr's propagated uncertainty on the 147-nm share (6.08 % kinetic, 5.58 % two-temperature). The first alone makes 61 % of the quadrature sum's square. The full list of 24 is Table 2 of `Kr_LIGHT/docs/accuracy_overview_v0_5.pdf`.

Numerical error is far below the physical uncertainty:

- The ledger closes to 4×10^{-15} (kinetic) and 2×10^{-14} (two-temp.).
- Halving the kinetic EEDF's renewal step moves the 147-nm share by 2×10^{-6} of itself; doubling its energy cells moves it by 1×10^{-3}.
- A tenfold tighter solver tolerance moves it by 8×10^{-8}.
- Time-stepping the afterglow's EEDF instead of solving it steady moves it by 2×10^{-4}.
- The two-temperature closure lies -8 % from the kinetic EEDF at the reference point and within 6 % at the scan's corners.

2.3 The reference point

percent of E_{dep} unless stated	kinetic EEDF	two-temperature
147 nm, Kr_2^*	6.08	5.58
near IR, 5p$\rightarrow$5s	0.48	0.48
heat, reactions and relaxation	90.69	90.67
heat, elastic collisions	3.23	3.75
Kr_2^* formed that radiates / is ionized / redissociates (%)	28.5 / 56.9 / 14.0	26.0 / 59.9 / 13.5
5s$\rightarrow$5p pumpings per Kr_2^* formed, net	25	24
T_e, pulse mean (eV)	1.57	1.53
n_e, pulse mean (cm^{-3})	8.2×10^{14}	8.6×10^{14}
T_g at the end (K)	856	859

Table 8: LIGHT_Kr's reference point at $t = 2$ µs (1 atm, 300 K, 300 kW cm^{-3} for 1 µs). The 5s$\leftrightarrow$5p cycle turns 73.3 % of the input into heat, 102 eV for each 147-nm photon. Over pressure and power the share runs from 0 at 0.1 atm to 19.85 % at 2 atm and 100 kW cm^{-3} (two-temperature closure).

2.4 The energy ladder

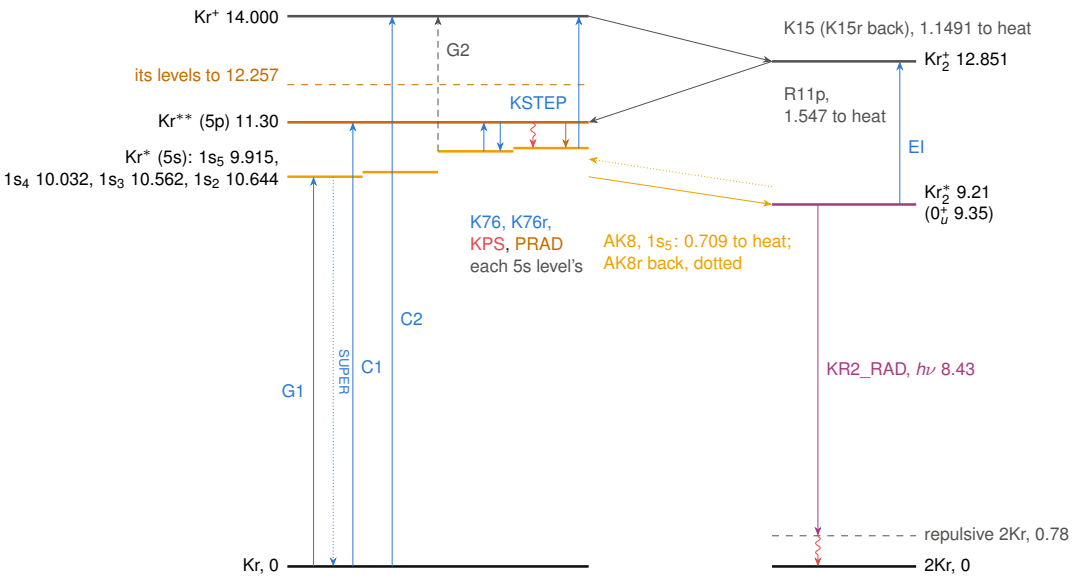


Figure 2: LIGHT_Kr v0_5's energy ladder in eV above ground-state krypton at rest; the four 5s levels are drawn side by side. Blue arrows are electron impact, which takes its energy from the electron gas; K76r returns it. KSTEP_S and KSTEP_P ionize the 5s levels (3.356–4.084 eV) and the 5p rung (2.696 eV from $2p_{10}$), EI ionizes Kr_2^* (3.644 eV), and SUPER (dotted) returns a 5s level to the ground state. The dashed G2 is Penning ionization of the 5s levels on one another. A red squiggle is heat: KPS relaxes the 5p rung on krypton, and PRAD radiates its gap in the near infrared. The excimer forms out of each 5s level, redissociates by detailed balance (AK8r), and radiates 8.43 eV at 147 nm onto the repulsive ground state. C1's levels, 11.30–13.60 eV, cascade to the 5p rung. MIX, the Hornbeck–Molnar band and the resonance lines are not drawn. From `Kr_LIGHT/docs/accuracy_overview_v0_5.pdf`, Figure 2.

species	E_{form} (eV)
Kr^+	14.000
Kr_2^+	12.851
Kr^{**}	11.303
$Kr^*(1s_2)$	10.644
$Kr^*(1s_3)$	10.562
$Kr^*(1s_4)$	10.032
$Kr^*(1s_5)$	9.915
$Kr_2^*(0_u^+)$	9.348
Kr_2^*	9.206
Kr	0.000

Table 9: The rungs of LIGHT_Kr's ladder as the code books them (`species.e_form`), above ground-state krypton.

2.5 Where LIGHT_Kr is not yet reliable

- The electrons' collisions with Kr_2^* decide the answer and are unmeasured. Its ionization is a calculation; its quenching and state mixing have neither measurement nor calculation (bands at 0, to Kannari's estimate).

- The excimer's formation: the $1s_5$ measurements spread $\times 2.5$, $1s_4$ rests on one measurement, $1s_3$ is a proxy, and the temperature law is measured at 200–400 K while the reference point ends at 856 K.
- The superelastic returns are 5–9 times Blagoev et al.'s thermal rates for $1s_5$ and $1s_4$ (they move the reference point by $-0.5\,\%$).
- The field-driven 5s$\leftrightarrow$5p cycle is unscored. The $J = 1$ levels use proxies, and the 5p rung's levels are assumed populated at the gas temperature.
- The e-beam efficiency at low pump is +29 % off Morozov et al., and no process in the model explains it.
- Not carried at all: Kr_3^+, Penning ionization between Kr^* and Kr_2^*, the other 5s transfers by electrons, and $Kr^+ + 2e^-$.

A LIGHT v0_5: every reaction rate

The 59 rows of the default network (`network.build()`, the rows a default run integrates), in the code's order, with each row's rate law and its rate constant evaluated by the model's own code (`Model._k`) at $T_g = 300$ K, the 1-atm fill and every band at its default. Electron rows are given for a Maxwellian at $T_e = 1$ and 3 eV; a heavy-particle row's k does not depend on T_e ("same"). The model's default closure is the kinetic EEDF solved during the run, so for electron rows these Maxwellian values are a reference, not what a run uses. The ion-ion rows include the Debye screening at $n_e = 10^{14}$ cm^{-3}. 11 rows are 0 by default: they are bands with no measurement. The source column is the start of each row's citation in `v0_5/network.py`; `v0_5/SOURCES.md` says how each source was read.

Table 10: LIGHT v0_5's reaction rates.

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
`G1`	$e^- + Kr \rightarrow e^- + Kr^*$	Maxwellian $\langle\sigma v\rangle$, Kr excitation `Kr(s)(9.915eV)`	3.44×10^{-13}	3.78×10^{-10}	cm^3/s		notes (1) e + Kr -> Kr* 5s[3/2]. Biagi-v7.1 'Kr(s)' = the whole 5s manifold (5s[3/2]1,2 and 5s'[1/2]0,1) lumped at 9.915 eV; k = <sigma v> over a Maxwellian (notes: Q = . . .
`G2`	$2Kr^* \rightarrow Kr^+ + Kr + e^-$	band kr_pen	5×10^{-10}	same	cm^3/s	kr_pen	notes (2) Kr* + Kr* 'associative ionization', 5.0e-10 cm3/s, Kannari 1985 Table I Penning block from his ref. 86, Leichner & Ericson, PRA 9, 251 (1974), with Kr+ + Kr + . . .
`G3`	$Kr^* + F_2 \rightarrow KrF(B) + F$	4.73×10^{-10}	4.73×10^{-10}	same	cm^3/s		Kr* + F2 -> KrF(B) + F: 62 % of the 90 % of 8.47e-10 cm3/s that forms KrF*. The rate is the mean of the first-hand 7.2 (Velazco, Kolts & Setser 1976 via COO-2807-2 Table . . .
`G4`	$KrF(B) \rightarrow Kr + F$	$0.966\,/\tau$, τ = 6.99 ns (tau_sp_s)	1.38×10^{8}	same	s^{-1}		KrF(B) -> Kr + F + hv, 248 nm (B-X): the 96.6 % share of B's radiative decay, tau_B = 6.99 ns in all (Kannari 1985 Table I p.4312, 1.43e8 1/s; Eden 1978 measured 6.8 +- . . .
`G5`	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	$1.9 \times 10^{-30} \times (T_g/300)^{-3.32}$	1.9×10^{-30}	same	cm^6/s		KrF(B) + 2Kr -> Kr2F* + Kr, 1.90e-30 cm6/s at 300 K, ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the KrF(B,C) . . .
`G6`	$e^- + Kr \rightarrow e^- + Kr$	elastic energy loss on the Kr momentum-transfer cross section; no density change	–	–	cm^3/s		notes (6) e + Kr elastic. Biagi-v7.1 momentum-transfer set; power to the gas (2m/M)(1 - Tg/Te) n_e N <sigma_m v eps>
`C1a`	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle\sigma v\rangle$, Kr excitation `Kr(p)(11.30eV)`	5.93×10^{-14}	1.48×10^{-10}	cm^3/s		Biagi-v7.1 5p group (11.30 eV). Not in the notes: without it the electrons lose ~half their inelastic power nowhere. v0_4: the product is the 5p rung Kr** (11.30 eV, the . . .
`C1b`	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle\sigma v\rangle$, Kr excitation `Kr(d+p)(11.998eV)`	5.84×10^{-14}	2.53×10^{-10}	cm^3/s		Biagi-v7.1 d+p group (11.998 eV); cascade to the 5p rung (Kr**) taken prompt, the excess heats the gas (v0_4: product Kr**, 0.698 eV to heat)
`C1c`	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle\sigma v\rangle$, Kr excitation `Kr*(12.75eV)`	7.73×10^{-15}	1.29×10^{-10}	cm^3/s		Biagi-v7.1 higher group (12.75 eV); cascade to the 5p rung (Kr**) taken prompt, the excess heats the gas (v0_4: product Kr**, 1.450 eV to heat)
`C2`	$e^- + Kr \rightarrow 2e^- + Kr^+$	Maxwellian $\langle\sigma v\rangle$, Kr ionization `Kr^+`	1.57×10^{-14}	4.14×10^{-10}	cm^3/s		Biagi-v7.1 ionization (13.996 eV): the notes' 'Kr+' output needs electron-impact ionization

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
C3	$Kr_2F^* \rightarrow 2Kr + F$	4.5×10^{6}	4.5×10^{6}	same	s^{-1}		Kr2F* -> 2Kr + F + hv: Xu/Gadomski/Setser 1993 p.2597, tau = 220 +- 20 ns (REBUILD K169). Band maximum 420 +- 35 nm, printed (Huestis, Marowsky & Tittel 1984 Table 6.1 . . .
G3C	$Kr^* + F_2 \rightarrow KrF(C) + F$	2.9×10^{-10}	2.9×10^{-10}	same	cm^3/s		Kr* + F2 -> KrF(C) + F: 38 % of the KrF* from 8.47e-10 cm3/s (yield 0.90, see G3). Yu & Setser, J. Phys. Chem. 94, 2934 (1990) Sec. III.A: 'the true kB/kC ratio will be . . .
G3X	$Kr^* + F_2 \rightarrow Kr + 2F$	8.47×10^{-11}	8.47×10^{-11}	same	cm^3/s		Kr* + F2 without KrF*: the 10 % the measured yield (0.90, see G3) leaves. The products are not measured, and 'no other emissions were observed from the F2 reactions' . . .
G4C	$KrF(C) \rightarrow Kr + F$	2×10^{7}	2×10^{7}	same	s^{-1}		KrF(C) -> Kr + F + hv (C-A). tau_C = 50 ns [41, 63], ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the KrF(B,C) . . .
G4A	$KrF(B) \rightarrow Kr + F$	$0.0338 / \tau$, $\tau = 6.99$ ns (tau_sp_s)	4.84×10^{6}	same	s^{-1}		KrF(B) -> Kr + F + hv (B-A), the 3.4 % branch of B's radiative decay: A(B-A)/A(B-X) = 0.035, calculated (Dunning & Hay 1978, quoted by Yu & Setser, JPC 94, 2934 (1990) . . .
BC	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	1.05×10^{-11}	1.05×10^{-11}	same	cm^3/s		B -> C mixing by Kr, k_BC = 1.05e-11 cm3/s [4.6e-12, 1.7e-11], ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the . . .
CB	$KrF(C) + Kr \rightarrow KrF(B) + Kr$	detailed balance: $k_f / K(T_g)$, $k_f = 1.05 \times 10^{-11}$, $K_{300} = 1.1$ over the KrF(B)–KrF(C) gap	9.55×10^{-12}	same	cm^3/s		C -> B by detailed balance, k_BC / K_eq(Tg). K_eq = 1.10 at 300 K [0.84, 1.38], ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md . . .
G5C	$KrF(C) + 2Kr \rightarrow Kr_2F^* + Kr$	$3.79 \times 10^{-31} \times (T_g/300)^{-3.32}$	3.79×10^{-31}	same	cm^6/s		KrF(C) + 2Kr -> Kr2F* + Kr, 3.79e-31 cm6/s at 300 K, ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the KrF(B,C) . . .
QF2	$KrF(B) + F_2 \rightarrow Kr + 3F$	4.75×10^{-10}	4.75×10^{-10}	same	cm^3/s		KrF(B) + F2, 4.75e-10 cm3/s, ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the KrF(B,C) block) (Eden's B-only . . .
QF2C	$KrF(C) + F_2 \rightarrow Kr + 3F$	3.04×10^{-10}	3.04×10^{-10}	same	cm^3/s		KrF(C) + F2, 3.04e-10 cm3/s, ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the KrF(B,C) block)
QKR	$KrF(B) + Kr \rightarrow 2Kr + F$	10^{-14}	10^{-14}	same	cm^3/s		KrF(B) + Kr, 1.0e-14 cm3/s, the lower bound of ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the KrF(B,C) block) . . .
QKRC	$KrF(C) + Kr \rightarrow 2Kr + F$	8.37×10^{-13}	8.37×10^{-13}	same	cm^3/s		KrF(C) + Kr, 8.37e-13 cm3/s, ML joint fit 2026-09-27 to Eden 1978, Gadomski 1992 and Yu & Setser 1990 (ML/README.md, Results for the KrF(B,C) block)
QE	$KrF(B) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8}	6.6×10^{-8}	6.6×10^{-8}	cm^3/s		KrF* + e, 6.6e-8 cm3/s: Trainor & Jacob's measured product k_e tau / k_att = 2.9e-7 s (Kushner, Hanson & Schneider APL 55, 2482 (1989), recombination factor 0.62) . . .
QEC	$KrF(C) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8}	6.6×10^{-8}	6.6×10^{-8}	cm^3/s		as QE for the C state

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
`Q2F`	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	band kr2f_f2	1.2×10^{-10}	same	cm^3/s	kr2f_f2	Kr2F* + F2, (1.2 +- 0.2)e-10 cm3/s measured, Xu 1993 Table I p.2594 (REBUILD keeps Kannari's 1.5e-10). Band kr2f_f2 carries the other measurements, up to Quigley & . . .
`Q2FF`	$Kr_2F^* + F \rightarrow 2Kr + 2F$	band kr2f_f	0	same	cm^3/s	kr2f_f	Kr2F* + F, no Kr measurement, so 0 unless a band run sets it (kr2f_f, 0 to 1.8e-10 cm3/s). Jancaitis, UCRL-53465 (1983) p.153 and Fig. 11, found Ar2F* quenched after the . . .
`K15`	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	$2.4 \times 10^{-31} \times (T_g/300)^{-0.75}$	2.4×10^{-31}	same	cm^6/s		Kr+ + 2Kr -> Kr2+ + Kr, 2.4e-31 cm6/s at 300 K, the mean of three measurements: 2.3e-31 (Bhattacharya, PRA 1, 1196 (1970)), (2.2 +- 0.2)e-31 (Grossl et al., JCP 74, 1728 . . .
`R11p`	$e^- + Kr_2^+ \rightarrow Kr^{**} + Kr$	$1.6 \times 10^{-6} \, (T_e/300\,K)^{-0.55}$	2.14×10^{-7}	1.17×10^{-7}	cm^3/s		Kr2+ + e -> Kr** + Kr, 1.6e-6 (300 K/Te)^0.55 cm3/s +- 10 %, Shiu & Biondi, PRA 16, 1817 (1977) (abstract, Te 300-8400 K, Tgas 300 K); Mikus, J. Phys. D 11, L39 (1978) . . .
`ATT`	$e^- + F_2 \rightarrow F^- + F$	Maxwellian $\langle \sigma v \rangle$, F2 attachment	3.51×10^{-9}	9.75×10^{-10}	cm^3/s		dissociative attachment, SIGLO set (the F2 file REBUILD reads)
`DIS_a3`	$e^- + F_2 \rightarrow e^- + 2F$	Maxwellian $\langle \sigma v \rangle$, F2 excitation `F2(a3)(3.16eV)`	1.13×10^{-10}	1.35×10^{-9}	cm^3/s		SIGLO a3Pi_u (repulsive: excitation = dissociation, REBUILD 0.1.14 reroute)
`DIS_A1`	$e^- + F_2 \rightarrow e^- + 2F$	Maxwellian $\langle \sigma v \rangle$, F2 excitation `F2(A1)(4.34eV)`	1.93×10^{-11}	4.02×10^{-10}	cm^3/s		SIGLO A1Pi_u (repulsive, as DIS_a3)
`VIB1`	$e^- + F_2 \rightarrow e^- + F_2$	Maxwellian $\langle \sigma v \rangle$, F2 excitation `F2(V1)(0.111eV)`	1.22×10^{-9}	4.27×10^{-10}	cm^3/s		SIGLO v=1, the quantum relaxes to heat
`VIB2`	$e^- + F_2 \rightarrow e^- + F_2$	Maxwellian $\langle \sigma v \rangle$, F2 excitation `F2(V2)(0.219eV)`	1.49×10^{-9}	5.47×10^{-10}	cm^3/s		SIGLO v=2
`VIB3`	$e^- + F_2 \rightarrow e^- + F_2$	Maxwellian $\langle \sigma v \rangle$, F2 excitation `F2(V3)(0.324eV)`	9.75×10^{-10}	3.78×10^{-10}	cm^3/s		SIGLO v=3
`VIB4`	$e^- + F_2 \rightarrow e^- + F_2$	Maxwellian $\langle \sigma v \rangle$, F2 excitation `F2(V4)(0.421eV)`	3.27×10^{-10}	1.31×10^{-10}	cm^3/s		SIGLO v=4
`K26`	$Kr^+ + F^- \rightarrow KrF(B)$	harmonic mean of $k_3 N (T_g/300)^{-3}$ and k_L/N, $k_3 = 4.82 \times 10^{-6}$, $k_L = 5.66 \times 10^{-6}$ (1 amagat), $\times$ Debye screening $\times$ ii_yield $\times(1-$ii_f_c)	1.79×10^{-6}	same	cm^3/s	ii_f_c, ii_yield	Kr+ + F- -> KrF(B), Flannery form of Kannari 1985 Sec. II.B (the modified Natanson calculation of Hoffman & Moreno, SAND80-1486, which Kannari used gas-temperature . . .
`K26C`	$Kr^+ + F^- \rightarrow KrF(C)$	harmonic mean of $k_3 N (T_g/300)^{-3}$ and k_L/N, $k_3 = 4.82 \times 10^{-6}$, $k_L = 5.66 \times 10^{-6}$ (1 amagat), $\times$ Debye screening $\times$ ii_yield $\times$ ii_f_c	0	same	cm^3/s	ii_f_c, ii_yield	Kr+ + F- -> KrF(C), the ii_f_c share of K26's KrF*
`K26X`	$Kr^+ + F^- \rightarrow Kr + F$	harmonic mean of $k_3 N (T_g/300)^{-3}$ and k_L/N, $k_3 = 4.82 \times 10^{-6}$, $k_L = 5.66 \times 10^{-6}$ (1 amagat), $\times$ Debye screening $\times(1-$ii_yield)	0	same	cm^3/s	ii_yield	Kr+ + F- -> Kr + F, the 1 - ii_yield share of K26 that forms no KrF*, its energy to heat
`K108`	$Kr_2^+ + F^- \rightarrow KrF(B) + Kr$	harmonic mean of $k_3 N (T_g/300)^{-3}$ and k_L/N, $k_3 = 4.83 \times 10^{-6}$, $k_L = 6.35 \times 10^{-6}$ (1 amagat), $\times$ Debye screening $\times$ ii_yield $\times(1-$ii_f_c)	1.87×10^{-6}	same	cm^3/s	ii_f_c, ii_yield	Kr2+ + F- -> KrF(B) + Kr, Flannery form of Kannari 1985 Sec. II.B (the modified Natanson calculation of Hoffman & Moreno, SAND80-1486, which Kannari used gas-temperature . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
K108C	$Kr_2^+ + F^- \rightarrow KrF(C) + Kr$	harmonic mean of $k_3 N (T_g/300)^{-3}$ and k_L/N, $k_3 = 4.83 \times 10^{-6}$, $k_L = 6.35 \times 10^{-6}$ (1 amagat), $\times$ Debye screening $\times$ ii_yield $\times$ ii_f_c	0	same	cm^3/s	ii_f_c, ii_yield	Kr2+ + F- -> KrF(C) + Kr, the ii_f_c share of K108
K108X	$Kr_2^+ + F^- \rightarrow 2Kr + F$	harmonic mean of $k_3 N (T_g/300)^{-3}$ and k_L/N, $k_3 = 4.83 \times 10^{-6}$, $k_L = 6.35 \times 10^{-6}$ (1 amagat), $\times$ Debye screening $\times(1-$ii_yield)	0	same	cm^3/s	ii_yield	Kr2+ + F- -> 2Kr + F, the 1 - ii_yield share of K108
FF	$2F + Kr \rightarrow F_2 + Kr$	2.8×10^{-34}	2.8×10^{-34}	same	cm^6/s		F + F + Kr -> F2 + Kr, 2.8e-34 cm6/s, the evaluation of Cohen & Bott (Aerospace TR-0076(6603)-2 (1976), DTIC ADA024332, pp.12-16: '10^13.7 exp(-35,100/RT) is . . .
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	band kr2_form $\times (T_g/300)^{-0.64}$	1.7×10^{-32}	same	cm^6/s	kr2_form	Kr*(3P2) + 2Kr -> Kr2* + Kr, 1.7 +- 0.2e-32 cm6/s at 300 K, Ueda, Kanada, Sasaki & Takuma, Rev. Laser Eng. 12, 357 (1984) Table V (e-beam, Kr* absorption at 0.5-1 atm . . .
KR2_RAD	$Kr_2^* \rightarrow 2Kr$	3.7×10^6	3.7×10^6	same	s^{-1}		Kr2* -> 2Kr + hv, 3.7e6 1/s = 270 ns, the slow (1u, 0u-) component that carries the kinetics: Morikawa et al., JCP 91, 1469 (1989) 270 +- 5 ns; Salamero 1979 250 +- 20 . . .
NK7	$Kr_2^* + F_2 \rightarrow KrF(B) + Kr + F$	band 0.5 $\times$ kr2_f2	1.05×10^{-10}	same	cm^3/s	kr2_f2	half of Kr2* + F2 = 2.1 +- 0.2e-10 cm3/s (Quigley & Hughes, APL 32, 649 (1978) p.650, read first-hand). Their products 'cannot be discerned in this experiment', and the . . .
K2F_BR	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	band 0.5 $\times$ kr2_f2	1.05×10^{-10}	same	cm^3/s	kr2_f2	the other half of Quigley & Hughes' 2.1e-10 (as NK7); J&H #20. Chen & Payne, IEEE JQE 15, 149 (1979) find this direct channel minor (Fig. 25, f2'/f3 << 1: Kr2F* forms . . .
K2F_F	$Kr_2^* + F \rightarrow KrF(B) + Kr$	band kr2_f	0	same	cm^3/s	kr2_f	Kr2* + F -> KrF(B) + Kr, no measurement, so 0 unless a band run sets it (kr2_f, 0 to 1e-9 cm3/s). Jancaitis's LLNL model (UCRL-53465, 1983, Table II pp.94-95) has no . . .
PEN_KR2	$2Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	band kr2_pen	0	same	cm^3/s	kr2_pen	Kr2* + Kr2* -> Kr2+ + 2Kr + e, no measurement, so 0 unless a band run sets it (kr2_pen, 0 to 5.0e-10 cm3/s). Jancaitis's model carries 1.0e-10 from Werner et al., IEEE . . .
KPS	$Kr^{**} + Kr \rightarrow Kr^* + Kr$	band kps_kr	1.7×10^{-10}	same	cm^3/s	kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr, the p -> s relaxation. Band kps_kr, default (1.7 +- 0.2)e-10 cm3/s: the one Kr measurement in hand, the total two-body loss of Kr 5p'[3/2]2 . . .
PRAD	$Kr^{**} \rightarrow Kr^*$	3.44×10^7	3.44×10^7	same	s^{-1}		Kr(5p) -> Kr(5s) + hv, the near-infrared 5p -> 5s lines (760-830 nm), which end on the 5s levels and are not trapped. One lifetime for the manifold, 29.0 ns, the . . .
K76	$e^- + Kr^* \rightarrow e^- + Kr^{**}$	Maxwellian rate of the lumped measured 5s$\rightarrow$5p σ (Jung et al.) $\times$ k_5s5p	2.78×10^{-7}	1.05×10^{-6}	cm^3/s	k_5s5p	e + Kr(5s) -> e + Kr(5p) on the measured cross sections out of the two metastables into the ten 5p levels (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) . . .
K76r	$e^- + Kr^{**} \rightarrow e^- + Kr^*$	Klein–Rosseland reverse of the lumped measured σ, g ratio 0.333 $\times$ k_5s5p	3.71×10^{-7}	5.54×10^{-7}	cm^3/s	k_5s5p	e + Kr(5p) -> e + Kr(5s), the reverse of K76 by the Klein-Rosseland relation on the same cross section, g(5s)/g(5p) = 12/36 over the ledger's gap; its 1.385 eV returns . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
KSTEP_S	$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	kstep_s $T_e^{0.68} e^{-4.04/T_e}$	0	0	cm^3/s	kstep_s	e + Kr(5s) -> Kr+ + 2e, stepwise ionization. Band kstep_s: no measurement; Kannari 1985 Table I [79] prints Vriens's binary-encounter calculation 7.0e-8 Te^0.68 . . .
KSTEP_P	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	kstep_p $T_e^{0.62} e^{-2.53/T_e}$	0	0	cm^3/s	kstep_p	e + Kr(5p) -> Kr+ + 2e. Band kstep_p: no measurement; Kannari 1985 Table I [79], Vriens's calculation 2.0e-7 Te^0.62 exp(-2.53/Te). The 2.70 eV comes from the electron . . .
K27	$Kr^{**} + F_2 \rightarrow KrF(B) + F$	band $0.558 \times$ k5p_f2	7.25×10^{-10}	same	cm^3/s	k5p_f2	Kr(5p) + F2 -> KrF(B) + F. Band k5p_f2 on the total Kr** + F2 rate, default (13 +- 2)e-10 cm3/s: the quenching of the collisionally coupled Kr(2p8, 2p9) pair by F2 to . . .
K27C	$Kr^{**} + F_2 \rightarrow KrF(C) + F$	band $0.342 \times$ k5p_f2	4.45×10^{-10}	same	cm^3/s	k5p_f2	Kr(5p) + F2 -> KrF(C) + F, the C share of K27 (as G3C)
K27X	$Kr^{**} + F_2 \rightarrow Kr + 2F$	band $0.1 \times$ k5p_f2	1.3×10^{-10}	same	cm^3/s	k5p_f2	Kr(5p) + F2 without KrF*, the 10 % (as G3X)
QF	$KrF(B) + F \rightarrow Kr + 2F$	band krf_f	0	same	cm^3/s	krf_f	KrF(B) + F -> Kr + 2F. Band krf_f, 0 by default: no measurement. Kannari 1985 Table I p.4311 prints 7.8e-10 cm3/s 'analogous to F2' (the F2 quench value he uses), the . . .
QFC	$KrF(C) + F \rightarrow Kr + 2F$	band krf_f	0	same	cm^3/s	krf_f	KrF(C) + F -> Kr + 2F. Band krf_f, 0 by default: no measurement. Kannari 1985 Table I p.4311 prints 7.8e-10 cm3/s 'analogous to F2' (the F2 quench value he uses), the . . .

Optional rows. The group `electron` (`-add electron`) adds these 2 rows; a default run leaves them out.

row	group	reaction	rate law	source (excerpt)
KSTEP	electron	$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	$1.56 \times 10^{-7} T_e^{0.71} e^{-E/T_e}$, $E = 4.09$ eV	Chen R9 template, E_bind 4.09 eV, EST; J&H #48 discharge value 1.5e-8 cm3/s agrees (REBUILD KSTEP)
SUPER	electron	$e^- + Kr^* \rightarrow e^- + Kr$	Klein–Rosseland reverse of `Kr(s)(9.915eV)`, g ratio 0.0833	detailed balance of G1, g(ground)/g(5s manifold) = 1/12 (statistical over the four 5s levels), EST — LIGHT's own row, REBUILD carries superelastics inside its swarm . . .

Table 12: LIGHT v0_5's bands (`network.BANDS`): rates carried with a default and two ends. A rate with no measurement defaults to 0 and its band runs to the largest published value. The propagated uncertainty runs each band's two ends.

band	rate	default	range	unit	rows that read it
kr2_f	Kr2* + F -> KrF(B) + Kr	0	0 to 10^{-9}	cm3/s	K2F_F
kr2_pen	Kr2* + Kr2* -> Kr2+ + 2Kr + e	0	0 to 5×10^{-10}	cm3/s	PEN_KR2
kr2f_f	Kr2F* + F -> 2Kr + 2F	0	0 to 1.8×10^{-10}	cm3/s	Q2FF
kr2_form	Kr* + 2Kr -> Kr2* + Kr at 300 K	1.7×10^{-32}	1.7×10^{-32} to 5.5×10^{-32}	cm6/s	AK8_Kr
kr2_f2	Kr2* + F2, both channels	2.1×10^{-10}	2.1×10^{-10} to 5×10^{-10}	cm3/s	NK7, K2F_BR
kr_pen	Kr* + Kr* -> Kr+ + Kr + e	5×10^{-10}	1.5×10^{-10} to 5×10^{-10}	cm3/s	G2

band	rate	default	range	unit	rows that read it
kr2f_f2	Kr2F* + F2 -> 2Kr + F + F2	1.2×10^{-10}	1.2×10^{-10} to 4.3×10^{-10}	cm3/s	`Q2F`
trimer_tg_pow	KrF* + 2Kr -> Kr2F* + Kr, its (Tg/300)^n	-3.32	-3.32 to -0.5	exponent	none (not a row rate)
krf_f	KrF(B,C) + F -> Kr + 2F	0	0 to 7.8×10^{-10}	cm3/s	`QF, QFC`
kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr	1.7×10^{-10}	3×10^{-11} to 1.7×10^{-10}	cm3/s	`KPS`
k_5s5p	e + Kr(5s) -> e + Kr(5p), a scale on the measured lumped cross section (jung2006.py)	1	0.5 to 1.4	factor	`K76, K76r`
kstep_s	e + Kr(5s) -> Kr+ + 2e, the pre-exponential of Kannari's Te^0.68 exp(-4.04/Te)	0	0 to 1.5×10^{-7}	cm3/s	`KSTEP_S`
kstep_p	e + Kr(5p) -> Kr+ + 2e, the pre-exponential of Kannari's Te^0.62 exp(-2.53/Te)	0	0 to 2×10^{-7}	cm3/s	`KSTEP_P`
k5p_f2	Kr(5p) + F2, all channels	1.3×10^{-9}	0 to 1.3×10^{-9}	cm3/s	`K27, K27C, K27X`
ii_yield	KrF* per ion-ion recombination (K26, K108)	1	0.5 to 1	fraction	`K26, K26C, K26X, K108, K108C, K108X`
ii_f_c	share of the ion-ion KrF* formed in C	0	0 to 0.5	fraction	`K26, K26C, K108, K108C`
ib_scale	the electron-ion (Turnbull) share of the EM wave's absorbed power, a factor	1	0.9 to 1.1	factor	none (not a row rate)

B LIGHT_Kr v0_5: every reaction rate

The 219 rows of the default network, in the code's order, evaluated as in Appendix A ($T_g = 300$ K, 1 atm, bands at their defaults; electron rows on a Maxwellian at 1 and 3 eV). 31 rows are 0 by default. A row marked † leaves from one level of the 5p rung: its k is per atom in that level, and the model multiplies it by the level's share of Kr** at T_g (98 % $2p_{10}$ at 300 K). The Kr excitation rows run on Biagi's Magboltz 8.97 cross sections with v0_3's trained factors. The source column is the start of each row's citation in `v0_5/network.py`; `v0_5/SOURCES.md` has the full sources.

Table 13: LIGHT_Kr v0_5's reaction rates.

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
`G1_1s5`	$e^- + Kr \rightarrow e^- + Kr^*(1s_5)$	Maxwellian $\langle \sigma v \rangle$, Kr excitation `Kr(1S5)`	1.34×10^{-13}	8.32×10^{-11}	cm^3/s		notes (1) e + Kr -> Kr* 5s[3/2]. Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(1S5) at 9.9152 eV: a 5s level, its own species since v0_4 (v0_1 to v0_3 lumped the four at the . . .
`G1_1s4`	$e^- + Kr \rightarrow e^- + Kr^*(1s_4)$	Maxwellian $\langle \sigma v \rangle$, Kr excitation `Kr(1S4)`	1.77×10^{-13}	1.55×10^{-10}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(1S4) at 10.0324 eV: a 5s level, its own species since v0_4 (v0_1 to v0_3 lumped the four at the 1s5 energy)
`G1_1s3`	$e^- + Kr \rightarrow e^- + Kr^*(1s_3)$	Maxwellian $\langle \sigma v \rangle$, Kr excitation `Kr(1S3)`	1.16×10^{-14}	1.24×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(1S3) at 10.5624 eV: a 5s level, its own species since v0_4 (v0_1 to v0_3 lumped the four at the 1s5 energy)
`G1_1s2`	$e^- + Kr \rightarrow e^- + Kr^*(1s_2)$	Maxwellian $\langle \sigma v \rangle$, Kr excitation `Kr(1S2)`	5.97×10^{-14}	9.53×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(1S2) at 10.6436 eV: a 5s level, its own species since v0_4 (v0_1 to v0_3 lumped the four at the 1s5 energy)

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
G2_1s5_1s5	$2Kr^*(1s_5) \rightarrow Kr^+ + Kr + e^-$	band kr_pen	5×10^{-10}	same	cm^3/s	kr_pen	Kr*(1s5) + Kr*(1s5) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s5_1s4	$Kr^*(1s_5) + Kr^*(1s_4) \rightarrow Kr^+ + Kr + e^-$	band $2 \times$ kr_pen	10^{-9}	same	cm^3/s	kr_pen	Kr*(1s5) + Kr*(1s4) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s5_1s3	$Kr^*(1s_5) + Kr^*(1s_3) \rightarrow Kr^+ + Kr + e^-$	band $2 \times$ kr_pen	10^{-9}	same	cm^3/s	kr_pen	Kr*(1s5) + Kr*(1s3) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s5_1s2	$Kr^*(1s_5) + Kr^*(1s_2) \rightarrow Kr^+ + Kr + e^-$	band $2 \times$ kr_pen	10^{-9}	same	cm^3/s	kr_pen	Kr*(1s5) + Kr*(1s2) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s4_1s4	$2Kr^*(1s_4) \rightarrow Kr^+ + Kr + e^-$	band kr_pen	5×10^{-10}	same	cm^3/s	kr_pen	Kr*(1s4) + Kr*(1s4) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s4_1s3	$Kr^*(1s_4) + Kr^*(1s_3) \rightarrow Kr^+ + Kr + e^-$	band $2 \times$ kr_pen	10^{-9}	same	cm^3/s	kr_pen	Kr*(1s4) + Kr*(1s3) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s4_1s2	$Kr^*(1s_4) + Kr^*(1s_2) \rightarrow Kr^+ + Kr + e^-$	band $2 \times$ kr_pen	10^{-9}	same	cm^3/s	kr_pen	Kr*(1s4) + Kr*(1s2) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s3_1s3	$2Kr^*(1s_3) \rightarrow Kr^+ + Kr + e^-$	band kr_pen	5×10^{-10}	same	cm^3/s	kr_pen	Kr*(1s3) + Kr*(1s3) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s3_1s2	$Kr^*(1s_3) + Kr^*(1s_2) \rightarrow Kr^+ + Kr + e^-$	band $2 \times$ kr_pen	10^{-9}	same	cm^3/s	kr_pen	Kr*(1s3) + Kr*(1s2) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G2_1s2_1s2	$2Kr^*(1s_2) \rightarrow Kr^+ + Kr + e^-$	band kr_pen	5×10^{-10}	same	cm^3/s	kr_pen	Kr*(1s2) + Kr*(1s2) -> Kr+ + Kr + e, the notes' (2) Kr* + Kr* 'associative ionization' on the band kr_pen (5.0e-10 cm3/s, Kannari 1985 Table I from Leichner & Ericson …
G6	$e^- + Kr \rightarrow e^- + Kr$	elastic energy loss on the Kr momentum-transfer cross section; no density change	–	–	cm^3/s		notes (6) e + Kr elastic. Biagi (Magboltz 8.97) momentum-transfer cross section; power to the gas (2m/M)(1 - Tg/Te) n_e N <sigma_m v eps>
C1a_2p10	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P10)	5.14×10^{-15}	1.3×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P10) at 11.3035 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by …
C1a_2p9	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P9)	1.09×10^{-14}	2.98×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P9) at 11.4430 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by …
C1a_2p8	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P8)	1.26×10^{-14}	3.15×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P8) at 11.4447 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by …
C1a_2p7	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P7)	5.38×10^{-15}	1.38×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P7) at 11.5261 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by …

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
C1a_2p6	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P6)	8.45×10^{-15}	1.67×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P6) at 11.5458 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by . . .
C1a_2p5	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P5)	6.72×10^{-15}	2.31×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P5) at 11.6660 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by . . .
C1a_2p4	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P4)	2.36×10^{-15}	1.06×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P4) at 12.1004 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by . . .
C1a_2p3	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P3)	2.2×10^{-15}	9.27×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P3) at 12.1404 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by . . .
C1a_2p2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P2)	4.94×10^{-15}	1.98×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P2) at 12.1437 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by . . .
C1a_2p1	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2P1)	1.68×10^{-15}	1.01×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2P1) at 12.2565 eV: a 5p level; Kr** is the 5p rung at the 2p10 energy, the excess over it heats the gas. It relaxes to Kr* by . . .
C1b_3d6	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D6)	6.33×10^{-16}	4.58×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D6) at 11.9981 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_3d5	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D5)	2×10^{-15}	1.39×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D5) at 12.0370 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_3d3	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D3)	2.25×10^{-15}	1.65×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D3) at 12.1117 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_3d4_1	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D4!)	2.3×10^{-15}	1.65×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D4!) at 12.1253 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_3d4	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D4)	2.21×10^{-15}	1.11×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D4) at 12.1785 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_3d1_2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D1!!)	1.1×10^{-15}	6.27×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D1!!) at 12.2580 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_3d1_1	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D1!)	1.28×10^{-15}	4.84×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D1!) at 12.2843 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_2s5	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2S5)	2.61×10^{-15}	8.42×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2S5) at 12.3522 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1b_3d2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3D2)	2.48×10^{-15}	1.65×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3D2) at 12.3546 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
C1b_2s4	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2S4)	4.49×10^{-15}	3.05×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2S4) at 12.3853 eV: a 4d or 6s level; the cascade to the 5p rung (Kr**) is taken prompt and the excess heats the gas
C1c_3p10	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3P10)	1.01×10^{-15}	6.25×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3P10) at 12.7564 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_3p9	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3P9)	1.45×10^{-15}	8.48×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3P9) at 12.7847 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_3p8	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3P8)	1.26×10^{-15}	7.6×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3P8) at 12.7854 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_3s1_2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3S1!!)	1.02×10^{-15}	1.05×10^{-11}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3S1!!) at 12.8034 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the ...
C1c_3p7	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3P7)	6.45×10^{-16}	3.27×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3P7) at 12.8092 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_3p6	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3P6)	9.72×10^{-16}	4.97×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3P6) at 12.8153 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_3s1_4	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3S1!!!!)	5.96×10^{-16}	6.48×10^{-12}	cm^3/s		Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3S1!!!! at 12.8253 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the ...
C1c_3s1_3	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3S1!!!) $\times(1-$hm_kr)	7.59×10^{-16}	7.33×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3S1!!!) at 12.8573 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the ...
C1c_3p5	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3P5) $\times(1-$hm_kr)	1.57×10^{-15}	9.75×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3P5) at 12.8648 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_4d5	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D5) $\times(1-$hm_kr)	2.36×10^{-16}	2.21×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D5) at 12.8698 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_4d6	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D6) $\times(1-$hm_kr)	1.47×10^{-16}	1.4×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D6) at 12.9035 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_4d4_1	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D4!) $\times(1-$hm_kr)	5.72×10^{-16}	5.03×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D4!) at 12.9725 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_3s1_1	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3S1!) $\times(1-$hm_kr)	6.3×10^{-16}	6.46×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3S1!) at 13.0044 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...
C1c_4d4	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D4) $\times(1-$hm_kr)	5.37×10^{-16}	3.52×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D4) at 13.0080 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess ...

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
C1c_4d3	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D3) $\times(1-$hm_kr$)$	3.83×10^{-16}	3.06×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D3) at 13.0192 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_2s3	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(2S3) $\times(1-$hm_kr$)$	1.26×10^{-16}	8.82×10^{-13}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(2S3) at 13.0297 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_3s2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3S2) $\times(1-$hm_kr$)$	1.47×10^{-16}	1.54×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3S2) at 13.0365 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_4d1_2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D1!!) $\times(1-$hm_kr$)$	2.81×10^{-16}	2.18×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D1!!) at 13.0386 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the . . .
C1c_4d1_1	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D1!) $\times(1-$hm_kr$)$	3.4×10^{-16}	2.38×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D1!) at 13.0442 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_3s5	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3S5) $\times(1-$hm_kr$)$	1.04×10^{-15}	7.42×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3S5) at 13.0986 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_4d2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4D2) $\times(1-$hm_kr$)$	1.26×10^{-15}	1.38×10^{-11}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4D2) at 13.0987 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_3s4	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(3S4) $\times(1-$hm_kr$)$	1.03×10^{-15}	1.14×10^{-11}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(3S4) at 13.1139 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_4f	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4F SUM $\times(1-$hm_kr$)$	10^{-15}	6.72×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4F SUM at 13.1400 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the . . .
C1c_5d5	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(5D5) $\times(1-$hm_kr$)$	1.47×10^{-17}	1.9×10^{-13}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(5D5) at 13.3501 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_5d2	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(5D2) $\times(1-$hm_kr$)$	3.97×10^{-16}	5.37×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(5D2) at 13.4224 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_4s4	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(4S4) $\times(1-$hm_kr$)$	1.81×10^{-16}	2.47×10^{-12}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(4S4) at 13.4365 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
C1c_high	$e^- + Kr \rightarrow e^- + Kr^{**}$	Maxwellian $\langle \sigma v \rangle$, Kr excitation Kr(HIGH) $\times(1-$hm_kr$)$	1.24×10^{-15}	1.89×10^{-11}	cm^3/s	hm_kr	Biagi (Magboltz 8.97, LXCat 'Biagi') Kr(HIGH) at 13.6000 eV: a level from 12.76 eV up (6p, 5d, 7s, ...); the cascade to the 5p rung (Kr**) is taken prompt and the excess . . .
HM_3s1_3	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(3S1!!!)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(3S1!!!) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 12.8573 eV, 0.007 eV above Kr2+ (species.py): the share hm_kr . . .
HM_3p5	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(3P5)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(3P5) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 12.8648 eV, 0.014 eV above Kr2+ (species.py): the share hm_kr of . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
HM_4d5	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D5)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D5) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 12.8698 eV, 0.019 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4d6	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D6)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D6) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 12.9035 eV, 0.053 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4d4_1	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D4!)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D4!) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 12.9725 eV, 0.122 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_3s1_1	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(3S1!)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(3S1!) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0044 eV, 0.154 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4d4	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D4)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D4) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0080 eV, 0.157 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4d3	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D3)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D3) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0192 eV, 0.169 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_2s3	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(2S3)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(2S3) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0297 eV, 0.179 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_3s2	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(3S2)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(3S2) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0365 eV, 0.186 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4d1_2	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D1!!)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D1!!) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0386 eV, 0.188 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4d1_1	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D1!)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D1!) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0442 eV, 0.194 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_3s5	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(3S5)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(3S5) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0986 eV, 0.248 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4d2	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4D2)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4D2) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.0987 eV, 0.248 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_3s4	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(3S4)'s $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(3S4) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.1139 eV, 0.263 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4f	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4F SUM's $\langle \sigma v \rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4F SUM + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.1400 eV, 0.289 eV above Kr2+ (species.py): the share hm_kr of . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
HM_5d5	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(5D5)'s $\langle\sigma v\rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(5D5) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.3501 eV, 0.500 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_5d2	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(5D2)'s $\langle\sigma v\rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(5D2) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.4224 eV, 0.572 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_4s4	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(4S4)'s $\langle\sigma v\rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(4S4) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.4365 eV, 0.586 eV above Kr2+ (species.py): the share hm_kr of . . .
HM_high	$e^- + 2Kr \rightarrow 2e^- + Kr_2^+$	Hornbeck–Molnar branch of Kr(HIGH)'s $\langle\sigma v\rangle \times$ hm_kr	0	0	cm^3/s	hm_kr	Kr(HIGH) + Kr -> Kr2+ + e, the Hornbeck-Molnar ionization of the level Biagi (Magboltz 8.97) excites at 13.6000 eV, 0.749 eV above Kr2+ (species.py): the share hm_kr of . . .
C2	$e^- + Kr \rightarrow 2e^- + Kr^+$	Maxwellian $\langle\sigma v\rangle$, Kr ionization Kr^+	1.55×10^{-14}	4.01×10^{-10}	cm^3/s		Biagi (Magboltz 8.97) ionization (13.9996 eV): the notes' 'Kr+' output needs electron-impact ionization
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	$2.4 \times 10^{-31} \times (T_g/300)^{-0.75}$	2.4×10^{-31}	same	cm^6/s		Kr+ + 2Kr -> Kr2+ + Kr, 2.4e-31 cm6/s at 300 K, the mean of three measurements: 2.3e-31 (Bhattacharya, PRA 1, 1196 (1970)), (2.2 +- 0.2)e-31 (Grossl et al., JCP 74, 1728 . . .
K15r	$Kr_2^+ + Kr \rightarrow Kr^+ + 2Kr$	reverse of $2.4 \times 10^{-31}(T_g/300)^{-0.75}$ by detailed balance on Kr_2^+'s well	2.03×10^{-27}	same	cm^3/s		Kr2+ + Kr -> Kr+ + 2Kr, the redissociation of the molecular ion (v0_4): K15's rate over the equilibrium constant of Kr+ + Kr <-> Kr2+ (thermo.py), on the measured D0 = . . .
R11p	$e^- + Kr_2^+ \rightarrow Kr^{**} + Kr$	$1.6 \times 10^{-6}\,(T_e/300\,K)^{-0.55} \times F(T_g)^x$, $x =$ dr_tg	2.14×10^{-7}	1.17×10^{-7}	cm^3/s	dr_tg	Kr2+ + e -> Kr** + Kr, 1.6e-6 (300 K/Te)^0.55 cm3/s +- 10 %, Shiu & Biondi, PRA 16, 1817 (1977) (abstract, Te 300-8400 K, Tgas 300 K); Mikus, J. Phys. D 11, L39 (1978) . . .
AK8_1s5	$Kr^*(1s_5) + 2Kr \rightarrow Kr_2^* + Kr$	band kr2_form $\times (T_g/300)^{-0.64}$	3.81×10^{-32}	same	cm^6/s	kr2_form	Kr*(1s5, 3P2) + 2Kr -> Kr2* + Kr, band kr2_form, default 3.81 +- 0.08e-32 cm6/s: Marchal, Sewraj, Gardou, Merbahi & Yousfi, 'Study of formation and decay of rare-gas . . .
AK8_1s4	$Kr^*(1s_4) + 2Kr \rightarrow Kr_2^* + Kr$	band kr2_form_1s4 $\times (T_g/300)^{-0.64}$ $\times$(1−kr2_fast)	9.24×10^{-32}	same	cm^6/s	kr2_fast, kr2_form_1s4	Kr*(1s4, 3P1) + 2Kr -> Kr2* + Kr, band kr2_form_1s4, default 9.24 +- 0.80e-32 cm6/s: Marchal et al. 2018 Table 2, k31 (pure Kr, selective excitation). Audouard, Laporte . . .
AK8f_1s4	$Kr^*(1s_4) + 2Kr \rightarrow Kr_2^*(0_u^+) + Kr$	band kr2_form_1s4 $\times (T_g/300)^{-0.64}$ $\times$ kr2_fast	0	same	cm^6/s	kr2_fast, kr2_form_1s4	Kr*(1s4) + 2Kr -> Kr2*(0u+) + Kr, the share kr2_fast of the 1s4 association that radiates from the fast B 1Sigma_u+ (0u+) state (band, 0 by default: all to the slow . . .
AK8_1s3	$Kr^*(1s_3) + 2Kr \rightarrow Kr_2^* + Kr$	band kr2_form_1s3 $\times (T_g/300)^{-0.64}$	3.81×10^{-32}	same	cm^6/s	kr2_form_1s3	Kr*(1s3, 3P0) + 2Kr -> Kr2* + Kr, band kr2_form_1s3. Kholin, L'dov, Ustinovskii & Zayarnyi, J. Russ. Laser Res. 36, 74 (2015) (abstract) measured Kr(3P0) + Kr + Ar -> . . .
AK8r_1s5	$Kr_2^* + Kr \rightarrow Kr^*(1s_5) + 2Kr$	reverse of kr2_form$(T_g/300)^{-0.64}$ by detailed balance on Kr_2^*'s well (kr2_d0)	6.62×10^{-21}	same	cm^3/s	kr2_d0, kr2_form	Kr2* + Kr -> Kr*(1s5) + 2Kr. The reverse by detailed balance (thermo.py): the rate over the equilibrium constant of the excimer's A state on its measured well (band . . .
AK8r_1s4	$Kr_2^* + Kr \rightarrow Kr^*(1s_4) + 2Kr$	reverse of kr2_form_1s4$(T_g/300)^{-0.64}$ by detailed balance on Kr_2^*'s well (kr2_d0) $\times$(1−kr2_fast)	1.04×10^{-22}	same	cm^3/s	kr2_d0, kr2_fast, kr2_form_1s4	Kr2* + Kr -> Kr*(1s4) + 2Kr, AK8_1s4's reverse. The reverse by detailed balance (thermo.py): the rate over the equilibrium constant of the excimer's A state on its . . .
AK8fr_1s4	$Kr_2^*(0_u^+) + Kr \rightarrow Kr^*(1s_4) + 2Kr$	reverse of kr2_form_1s4$(T_g/300)^{-0.64}$ by detailed balance on $Kr_2^*(0_u^+)$'s well $\times$ kr2_fast	0	same	cm^3/s	kr2_fast, kr2_form_1s4	Kr2*(0u+) + Kr -> Kr*(1s4) + 2Kr, AK8f_1s4's reverse by detailed balance on the B state's measured well (0.684 eV, LaRocque et al. 1986)

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
AK8r_1s3	$Kr_2^* + Kr \rightarrow Kr^*(1s_3) + 2Kr$	reverse of kr2_form_1s3$(T_g/300)^{-0.64}$ by detailed balance on Kr_2^*'s well (kr2_d0)	1.78×10^{-32}	same	cm^3/s	kr2_d0, kr2_form_1s3	Kr2* + Kr -> Kr*(1s3) + 2Kr, AK8_1s3's reverse. The reverse by detailed balance (thermo.py): the rate over the equilibrium constant of the excimer's A state on its …
KR2_RAD	$Kr_2^* \rightarrow 2Kr$	3.7×10^6	3.7×10^6	same	s^{-1}		Kr2* -> 2Kr + hv, 3.7e6 1/s = 270 ns, the slow (1u, 0u-) component that carries the kinetics: Morikawa et al., JCP 91, 1469 (1989) 270 +- 5 ns; Salamero 1979 250 +- 20 …
KR2F_RAD	$Kr_2^*(0_u^+) \rightarrow 2Kr$	2.94×10^8	2.94×10^8	same	s^{-1}		Kr2*(0u+) -> 2Kr + hv, the fast (0u+) component of the second continuum, 3.4 +- 0.1 ns (Morikawa et al. 1989, the low-density limit; abstract)
QKR2	$Kr_2^* + Kr \rightarrow 3Kr$	band kr2_quench	0	same	cm^3/s	kr2_quench	Kr2* + Kr -> 3Kr, collisional quenching of the slow excimer: band kr2_quench, 0 by default (Oka, Rama Rao, Redpath & Firestone, JCP 61, 4740 (1974), 200-1400 Torr: 'no …
MIX_1s4	$Kr^*(1s_4) + Kr \rightarrow Kr^*(1s_5) + Kr$	$1.27 \times 10^{-13} \times (T_g/300)^{-1.29}$	1.27×10^{-13}	same	cm^3/s		Kr*(1s4) + Kr -> Kr*(1s5) + Kr, 1.3e-13 cm3/s at 295 K: Turner, Phys. Rev. 158, 121 (1967) (abstract: 13e-14 at 295 K, 22e-14 at 196 K, from the decay of the 123.6- and …
MIX_1s5r	$Kr^*(1s_5) + Kr \rightarrow Kr^*(1s_4) + Kr$	reverse of $1.27 \times 10^{-13}(T_g/300)^{-1.29}$ by detailed balance, g ratio 3	8.2×10^{-16}	same	cm^3/s		Kr*(1s5) + Kr -> Kr*(1s4) + Kr, MIX_1s4's reverse by detailed balance at the gas temperature, g(1s4)/g(1s5) exp(-0.117 eV/kTg); the gap comes from the gas
MIX_1s2	$Kr^*(1s_2) + 2Kr \rightarrow Kr^*(1s_4) + 2Kr$	2×10^{-33}	2×10^{-33}	same	cm^6/s		Kr*(1s2, 1P1) + 2Kr -> Kr*(1s4, 3P1) + 2Kr, about 0.2e-32 cm6/s: Audouard, Laporte, Subtil & Reininger, JCP 95, 3283 (1991) (abstract: above 4e20 cm^-3 the 1P1 decay …
MIX_1s4r2	$Kr^*(1s_4) + 2Kr \rightarrow Kr^*(1s_2) + 2Kr$	reverse of $2 \times 10^{-33}(T_g/300)^0$ by detailed balance, g ratio 3	1.08×10^{-43}	same	cm^6/s		Kr*(1s4) + 2Kr -> Kr*(1s2) + 2Kr, MIX_1s2's reverse by detailed balance, g(1s2)/g(1s4) exp(-0.611 eV/kTg)
RES_1s4	$Kr^*(1s_4) \rightarrow Kr$	band res_escape	0	same	s^{-1}	res_escape	Kr*(1s4) -> Kr + hv, 123.6 nm, the resonance radiation that escapes the trapping (A = 2.98e8 /s untrapped, NIST ASD): band res_escape, 0 by default (a 0-D box has no …
RES_1s2	$Kr^*(1s_2) \rightarrow Kr$	band res_escape	0	same	s^{-1}	res_escape	Kr*(1s2) -> Kr + hv, 116.5 nm, the escaping resonance radiation (A = 3.09e8 /s untrapped, NIST ASD), band res_escape as RES_1s4's
PEN_KR2	$2Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	band kr2_pen	0	same	cm^3/s	kr2_pen	Kr2* + Kr2* -> Kr2+ + 2Kr + e, no measurement, so 0 unless a band run sets it (kr2_pen, 0 to 5.0e-10 cm3/s). Jancaitis's model carries 1.0e-10 from Werner et al., IEEE …
EI_KR2	$e^- + Kr_2^* \rightarrow Kr_2^+ + 2e^-$	Vriens–Smeets σ, $E_n = 3.64$ eV, $\times$ ei_kr2	3.83×10^{-9}	5.86×10^{-8}	cm^3/s	ei_kr2	e + Kr2* -> Kr2+ + 2e, not carried before v0_5: Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in the kinetic EEDF …
EI_KR2F	$e^- + Kr_2^*(0_u^+) \rightarrow Kr_2^+ + 2e^-$	Vriens–Smeets σ, $E_n = 3.5$ eV, $\times$ ei_kr2	4.74×10^{-9}	6.58×10^{-8}	cm^3/s	ei_kr2	e + Kr2*(0u+) -> Kr2+ + 2e, as EI_KR2 at the fast state's ionization energy, 3.502 eV
EQ_KR2	$e^- + Kr_2^* \rightarrow e^- + 2Kr$	band eq_kr2	0	0	cm^3/s	eq_kr2	e + Kr2* -> e + 2Kr, the excimer's superelastic quenching: band eq_kr2, 0 by default (no measurement or calculation for any rare-gas excimer), to Kannari 1985 Table I's …

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
EM_KR2	$e^- + Kr_2^* \rightarrow e^- + Kr_2^*(0_u^+)$	em_kr2 $T_e^0 e^{-0.142/T_e}$	0	0	cm^3/s	em_kr2	e + Kr2* -> e + Kr2*(0u+), the electrons' mixing of the excimer's slow (1u, 0u-) and fast states: band em_kr2, 0 by default (no value for any rare-gas excimer), A . . .
EM_KR2r	$e^- + Kr_2^*(0_u^+) \rightarrow e^- + Kr_2^*$	reverse by detailed balance (g 3) of em_kr2 $T_e^0 e^{-0.142/T_e}$	0	0	cm^3/s	em_kr2	e + Kr2*(0u+) -> e + Kr2*, EM_KR2's reverse by detailed balance, g(1u, 0u-)/g(0u+) = 3
KPS_1s5	$Kr^{**} + Kr \rightarrow Kr^*(1s_5) + Kr$	band 0.417 × kps_kr × kps_5s	7.08×10^{-11}	same	cm^3/s	kps_5s, kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr, the p -> s relaxation. Band kps_kr, default (1.7 +- 0.2)e-10 cm3/s: the one Kr measurement in hand, the total two-body loss of Kr 5p'[3/2]2 . . .
KPS_1s4	$Kr^{**} + Kr \rightarrow Kr^*(1s_4) + Kr$	band 0.25 × kps_kr × kps_5s	4.25×10^{-11}	same	cm^3/s	kps_5s, kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr, the p -> s relaxation. Band kps_kr, default (1.7 +- 0.2)e-10 cm3/s: the one Kr measurement in hand, the total two-body loss of Kr 5p'[3/2]2 . . .
KPS_1s3	$Kr^{**} + Kr \rightarrow Kr^*(1s_3) + Kr$	band 0.0833 × kps_kr × kps_5s	1.42×10^{-11}	same	cm^3/s	kps_5s, kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr, the p -> s relaxation. Band kps_kr, default (1.7 +- 0.2)e-10 cm3/s: the one Kr measurement in hand, the total two-body loss of Kr 5p'[3/2]2 . . .
KPS_1s2	$Kr^{**} + Kr \rightarrow Kr^*(1s_2) + Kr$	band 0.25 × kps_kr × kps_5s	4.25×10^{-11}	same	cm^3/s	kps_5s, kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr, the p -> s relaxation. Band kps_kr, default (1.7 +- 0.2)e-10 cm3/s: the one Kr measurement in hand, the total two-body loss of Kr 5p'[3/2]2 . . .
KPS_low	$Kr^{**} + Kr \rightarrow Kr^*(1s_5) + Kr$	band kps_kr ×(1−kps_5s)	0	same	cm^3/s	kps_5s, kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr, the p -> s relaxation. Band kps_kr, default (1.7 +- 0.2)e-10 cm3/s: the one Kr measurement in hand, the total two-body loss of Kr 5p'[3/2]2 . . .
KPSr_1s5	$Kr^*(1s_5) + Kr \rightarrow Kr^{**} + Kr$	reverse of 0.417 × kps_kr by detailed balance at T_g (5p levels at T_g, g_{5s} = 5) × kps_5s	2.06×10^{-34}	same	cm^3/s	kps_5s, kps_kr	Kr*(1s5) + Kr -> Kr(5p) + Kr, KPS_1s5's reverse (v0_5), by detailed balance at the gas temperature with the rung's levels populated at Tg: k_f Z(Tg)/g(5s level) . . .
KPSr_1s4	$Kr^*(1s_4) + Kr \rightarrow Kr^{**} + Kr$	reverse of 0.25 × kps_kr by detailed balance at T_g (5p levels at T_g, g_{5s} = 3) × kps_5s	1.92×10^{-32}	same	cm^3/s	kps_5s, kps_kr	Kr*(1s4) + Kr -> Kr(5p) + Kr, KPS_1s4's reverse (v0_5), by detailed balance at the gas temperature with the rung's levels populated at Tg: k_f Z(Tg)/g(5s level) . . .
KPSr_1s3	$Kr^*(1s_3) + Kr \rightarrow Kr^{**} + Kr$	reverse of 0.0833 × kps_kr by detailed balance at T_g (5p levels at T_g, g_{5s} = 1) × kps_5s	1.54×10^{-23}	same	cm^3/s	kps_5s, kps_kr	Kr*(1s3) + Kr -> Kr(5p) + Kr, KPS_1s3's reverse (v0_5), by detailed balance at the gas temperature with the rung's levels populated at Tg: k_f Z(Tg)/g(5s level) . . .
KPSr_1s2	$Kr^*(1s_2) + Kr \rightarrow Kr^{**} + Kr$	reverse of 0.25 × kps_kr by detailed balance at T_g (5p levels at T_g, g_{5s} = 3) × kps_5s	3.55×10^{-22}	same	cm^3/s	kps_5s, kps_kr	Kr*(1s2) + Kr -> Kr(5p) + Kr, KPS_1s2's reverse (v0_5), by detailed balance at the gas temperature with the rung's levels populated at Tg: k_f Z(Tg)/g(5s level) . . .
KPSr_low	$Kr^*(1s_5) + Kr \rightarrow Kr^{**} + Kr$	reverse of kps_kr by detailed balance at T_g (5p levels at T_g, g_{5s} = 5) ×(1−kps_5s)	0	same	cm^3/s	kps_5s, kps_kr	Kr*(1s5) + Kr -> Kr(5p) + Kr, KPS_low's reverse (v0_5), by detailed balance at the gas temperature with the rung's levels populated at Tg: k_f Z(Tg)/g(5s level) . . .
PRAD_2p10_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	7.4×10^4; per atom in 2p10[†]	7.4×10^4	same	s^{-1}		Kr(2p10) -> Kr*(1s2) + hv, 1878.8 nm, A = 7.400e+04 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
PRAD_2p10_1s3	$Kr^{**} \rightarrow Kr^*(1s_3)$	1.26×10^5; per atom in 2p10†	1.26×10^5	same	s^{-1}		Kr(2p10) -> Kr*(1s3) + hv, 1673.0 nm, A = 1.260e+05 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .
PRAD_2p10_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	3.13×10^6; per atom in 2p10†	3.13×10^6	same	s^{-1}		Kr(2p10) -> Kr*(1s4) + hv, 975.4 nm, A = 3.130e+06 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .
PRAD_2p10_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	2.29×10^7; per atom in 2p10†	2.29×10^7	same	s^{-1}		Kr(2p10) -> Kr*(1s5) + hv, 893.1 nm, A = 2.289e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .
PRAD_2p9_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	3.61×10^7; per atom in 2p9†	3.61×10^7	same	s^{-1}		Kr(2p9) -> Kr*(1s5) + hv, 811.5 nm, A = 3.610e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p8_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	8.1×10^4; per atom in 2p8†	8.1×10^4	same	s^{-1}		Kr(2p8) -> Kr*(1s2) + hv, 1547.7 nm, A = 8.100e+04 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .
PRAD_2p8_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	2.22×10^7; per atom in 2p8†	2.22×10^7	same	s^{-1}		Kr(2p8) -> Kr*(1s4) + hv, 877.9 nm, A = 2.217e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p8_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	8.96×10^6; per atom in 2p8†	8.96×10^6	same	s^{-1}		Kr(2p8) -> Kr*(1s5) + hv, 810.6 nm, A = 8.960e+06 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p7_1s3	$Kr^{**} \rightarrow Kr^*(1s_3)$	7.6×10^4; per atom in 2p7†	7.6×10^4	same	s^{-1}		Kr(2p7) -> Kr*(1s3) + hv, 1286.5 nm, A = 7.600e+04 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .
PRAD_2p7_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	2.93×10^7; per atom in 2p7†	2.93×10^7	same	s^{-1}		Kr(2p7) -> Kr*(1s4) + hv, 830.0 nm, A = 2.931e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p7_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	4.27×10^6; per atom in 2p7†	4.27×10^6	same	s^{-1}		Kr(2p7) -> Kr*(1s5) + hv, 769.7 nm, A = 4.270e+06 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p6_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	3.1×10^5; per atom in 2p6†	3.1×10^5	same	s^{-1}		Kr(2p6) -> Kr*(1s2) + hv, 1374.2 nm, A = 3.100e+05 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
PRAD_2p6_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	8.94×10^6; per atom in 2p6†	8.94×10^6	same	s^{-1}		Kr(2p6) -> Kr*(1s4) + hv, 819.2 nm, A = 8.940e+06 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p6_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	2.73×10^7; per atom in 2p6†	2.73×10^7	same	s^{-1}		Kr(2p6) -> Kr*(1s5) + hv, 760.4 nm, A = 2.732e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p5_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	1.5×10^5; per atom in 2p5†	1.5×10^5	same	s^{-1}		Kr(2p5) -> Kr*(1s2) + hv, 1212.7 nm, A = 1.500e+05 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from . . .
PRAD_2p5_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	4.31×10^7; per atom in 2p5†	4.31×10^7	same	s^{-1}		Kr(2p5) -> Kr*(1s4) + hv, 759.0 nm, A = 4.310e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p4_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	1.81×10^7; per atom in 2p4†	1.81×10^7	same	s^{-1}		Kr(2p4) -> Kr*(1s2) + hv, 851.1 nm, A = 1.811e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p4_1s3	$Kr^{**} \rightarrow Kr^*(1s_3)$	1.58×10^7; per atom in 2p4†	1.58×10^7	same	s^{-1}		Kr(2p4) -> Kr*(1s3) + hv, 806.1 nm, A = 1.583e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p4_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	5×10^4; per atom in 2p4†	5×10^4	same	s^{-1}		Kr(2p4) -> Kr*(1s4) + hv, 599.5 nm, A = 5.000e+04 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p4_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	1.5×10^4; per atom in 2p4†	1.5×10^4	same	s^{-1}		Kr(2p4) -> Kr*(1s5) + hv, 567.4 nm, A = 1.500e+04 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p3_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	1.42×10^7; per atom in 2p3†	1.42×10^7	same	s^{-1}		Kr(2p3) -> Kr*(1s2) + hv, 828.3 nm, A = 1.418e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p3_1s3	$Kr^{**} \rightarrow Kr^*(1s_3)$	2.04×10^7; per atom in 2p3†	2.04×10^7	same	s^{-1}		Kr(2p3) -> Kr*(1s3) + hv, 785.7 nm, A = 2.041e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p3_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	5.5×10^4; per atom in 2p3†	5.5×10^4	same	s^{-1}		Kr(2p3) -> Kr*(1s4) + hv, 588.2 nm, A = 5.500e+04 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
PRAD_2p3_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	9.8×10^5; per atom in 2p3†	9.8×10^5	same	s^{-1}		Kr(2p3) -> Kr*(1s5) + hv, 557.2 nm, A = 9.800e+05 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p2_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	3.42×10^7; per atom in 2p2†	3.42×10^7	same	s^{-1}		Kr(2p2) -> Kr*(1s2) + hv, 826.5 nm, A = 3.416e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p2_1s4	$Kr^{**} \rightarrow Kr^*(1s_4)$	7.1×10^5; per atom in 2p2†	7.1×10^5	same	s^{-1}		Kr(2p2) -> Kr*(1s4) + hv, 587.2 nm, A = 7.100e+05 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p2_1s5	$Kr^{**} \rightarrow Kr^*(1s_5)$	1.1×10^5; per atom in 2p2†	1.1×10^5	same	s^{-1}		Kr(2p2) -> Kr*(1s5) + hv, 556.4 nm, A = 1.100e+05 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
PRAD_2p1_1s2	$Kr^{**} \rightarrow Kr^*(1s_2)$	4.06×10^7; per atom in 2p1†	4.06×10^7	same	s^{-1}		Kr(2p1) -> Kr*(1s2) + hv, 768.7 nm, A = 4.064e+07 /s (NIST ASD, Kramida, Ralchenko, Reader & NIST ASD Team, doi:10.18434/T4W30F, Kr I, read 2026-10-15 and -16), from the . . .
K76_1s5_2p2	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p2 × k_5s5p	9.54×10^{-10}	2.91×10^{-9}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p2) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale . . .
K76r_1s5_2p2	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p2 × k_5s5p; per atom in 2p2†	8.86×10^{-9}	6.13×10^{-9}	cm^3/s	k_5s5p	e + Kr(2p2) -> e + Kr*(1s5), the reverse of K76_1s5_2p2 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p2) = 5/5, out of the rung's level 2p2 at Tg . . .
K76_1s5_2p3	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p3 × k_5s5p	8.02×10^{-10}	2.59×10^{-9}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p3) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale . . .
K76r_1s5_2p3	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p3 × k_5s5p; per atom in 2p3†	1.24×10^{-8}	9.06×10^{-9}	cm^3/s	k_5s5p	e + Kr(2p3) -> e + Kr*(1s5), the reverse of K76_1s5_2p3 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p3) = 5/3, out of the rung's level 2p3 at Tg . . .
K76_1s5_2p4	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p4 × k_5s5p	7.13×10^{-10}	1.66×10^{-9}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p4) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale . . .
K76r_1s5_2p4	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p4 × k_5s5p; per atom in 2p4†	1.06×10^{-8}	5.74×10^{-9}	cm^3/s	k_5s5p	e + Kr(2p4) -> e + Kr*(1s5), the reverse of K76_1s5_2p4 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p4) = 5/3, out of the rung's level 2p4 at Tg . . .
K76_1s5_2p5	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p5 × k_5s5p	1.34×10^{-9}	2.38×10^{-9}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p5) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale . . .
K76r_1s5_2p5	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p5 × k_5s5p; per atom in 2p5†	3.86×10^{-8}	2.13×10^{-8}	cm^3/s	k_5s5p	e + Kr(2p5) -> e + Kr*(1s5), the reverse of K76_1s5_2p5 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p5) = 5/1, out of the rung's level 2p5 at Tg . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
K76_1s5_2p6	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p6 × k_5s5p	4.99×10^{-8}	2.11×10^{-7}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p6) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
K76r_1s5_2p6	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p6 × k_5s5p; per atom in 2p6†	2.55×10^{-7}	3.63×10^{-7}	cm^3/s	k_5s5p	e + Kr(2p6) -> e + Kr*(1s5), the reverse of K76_1s5_2p6 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p6) = 5/5, out of the rung's level 2p6 at Tg …
K76_1s5_2p7	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p7 × k_5s5p	1.17×10^{-8}	2.84×10^{-8}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p7) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
K76r_1s5_2p7	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p7 × k_5s5p; per atom in 2p7†	9.78×10^{-8}	8.09×10^{-8}	cm^3/s	k_5s5p	e + Kr(2p7) -> e + Kr*(1s5), the reverse of K76_1s5_2p7 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p7) = 5/3, out of the rung's level 2p7 at Tg …
K76_1s5_2p8	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p8 × k_5s5p	2.28×10^{-8}	8.52×10^{-8}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p8) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
K76r_1s5_2p8	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p8 × k_5s5p; per atom in 2p8†	1.05×10^{-7}	1.42×10^{-7}	cm^3/s	k_5s5p	e + Kr(2p8) -> e + Kr*(1s5), the reverse of K76_1s5_2p8 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p8) = 5/5, out of the rung's level 2p8 at Tg …
K76_1s5_2p9	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p9 × k_5s5p	1.29×10^{-7}	4.83×10^{-7}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p9) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
K76r_1s5_2p9	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p9 × k_5s5p; per atom in 2p9†	4.26×10^{-7}	5.74×10^{-7}	cm^3/s	k_5s5p	e + Kr(2p9) -> e + Kr*(1s5), the reverse of K76_1s5_2p9 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p9) = 5/7, out of the rung's level 2p9 at Tg …
K76_1s5_2p10	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s5→2p10 × k_5s5p	4.33×10^{-8}	1.56×10^{-7}	cm^3/s	k_5s5p	e + Kr*(1s5) -> e + Kr(2p10) on the measured cross section out of 1s5 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
K76r_1s5_2p10	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_5)$	Klein–Rosseland reverse of the measured 1s5→2p10 × k_5s5p; per atom in 2p10†	2.9×10^{-7}	4.14×10^{-7}	cm^3/s	k_5s5p	e + Kr(2p10) -> e + Kr*(1s5), the reverse of K76_1s5_2p10 by the Klein-Rosseland relation on the same cross section, g(1s5)/g(2p10) = 5/3, out of the rung's level 2p10 …
K76_1s4_2p2	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p2 × k_5s5p × k_5s5p_j1	1.04×10^{-9}	2.95×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p2) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set …
K76r_1s4_2p2	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p2 × k_5s5p × k_5s5p_j1; per atom in 2p2†	5.14×10^{-9}	3.58×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p2) -> e + Kr*(1s4), the reverse of K76_1s4_2p2 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p2) = 3/5, out of the rung's level 2p2 at Tg …
K76_1s4_2p3	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p3 × k_5s5p × k_5s5p_j1	8.7×10^{-10}	2.62×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p3) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set …
K76r_1s4_2p3	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p3 × k_5s5p × k_5s5p_j1; per atom in 2p3†	7.17×10^{-9}	5.3×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p3) -> e + Kr*(1s4), the reverse of K76_1s4_2p3 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p3) = 3/3, out of the rung's level 2p3 at Tg …

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
K76_1s4_2p4	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p4 × k_5s5p × k_5s5p_j1	7.73×10^{-10}	1.68×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p4) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set . . .
K76r_1s4_2p4	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p4 × k_5s5p × k_5s5p_j1; per atom in 2p4†	6.11×10^{-9}	3.34×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p4) -> e + Kr*(1s4), the reverse of K76_1s4_2p4 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p4) = 3/3, out of the rung's level 2p4 at Tg . . .
K76_1s4_2p5	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p5 × k_5s5p × k_5s5p_j1	1.44×10^{-9}	2.39×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p5) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set . . .
K76r_1s4_2p5	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p5 × k_5s5p × k_5s5p_j1; per atom in 2p5†	2.22×10^{-8}	1.23×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p5) -> e + Kr*(1s4), the reverse of K76_1s4_2p5 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p5) = 3/1, out of the rung's level 2p5 at Tg . . .
K76_1s4_2p6	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p6 × k_5s5p × k_5s5p_j1	5.41×10^{-8}	2.15×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p6) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set . . .
K76r_1s4_2p6	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p6 × k_5s5p × k_5s5p_j1; per atom in 2p6†	1.48×10^{-7}	2.13×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p6) -> e + Kr*(1s4), the reverse of K76_1s4_2p6 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p6) = 3/5, out of the rung's level 2p6 at Tg . . .
K76_1s4_2p7	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p7 × k_5s5p × k_5s5p_j1	1.26×10^{-8}	2.87×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p7) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set . . .
K76r_1s4_2p7	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p7 × k_5s5p × k_5s5p_j1; per atom in 2p7†	5.61×10^{-8}	4.72×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p7) -> e + Kr*(1s4), the reverse of K76_1s4_2p7 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p7) = 3/3, out of the rung's level 2p7 at Tg . . .
K76_1s4_2p8	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p8 × k_5s5p × k_5s5p_j1	2.46×10^{-8}	8.65×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p8) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set . . .
K76r_1s4_2p8	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p8 × k_5s5p × k_5s5p_j1; per atom in 2p8†	6.07×10^{-8}	8.31×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p8) -> e + Kr*(1s4), the reverse of K76_1s4_2p8 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p8) = 3/5, out of the rung's level 2p8 at Tg . . .
K76_1s4_2p9	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p9 × k_5s5p × k_5s5p_j1	1.4×10^{-7}	4.9×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p9) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set . . .
K76r_1s4_2p9	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p9 × k_5s5p × k_5s5p_j1; per atom in 2p9†	2.46×10^{-7}	3.36×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p9) -> e + Kr*(1s4), the reverse of K76_1s4_2p9 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p9) = 3/7, out of the rung's level 2p9 at Tg . . .
K76_1s4_2p10	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s4→2p10 × k_5s5p × k_5s5p_j1	4.68×10^{-8}	1.59×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s4) -> e + Kr(2p10) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set . . .
K76r_1s4_2p10	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_4)$	Klein–Rosseland reverse of the measured 1s4→2p10 × k_5s5p × k_5s5p_j1; per atom in 2p10†	1.67×10^{-7}	2.42×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p10) -> e + Kr*(1s4), the reverse of K76_1s4_2p10 by the Klein-Rosseland relation on the same cross section, g(1s4)/g(2p10) = 3/3, out of the rung's level 2p10 . . .

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
`K76_1s3_2p1`	$e^- + Kr^*(1s_3) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s3→2p1 × k_5s5p	3.13×10^{-9}	6.9×10^{-9}	cm^3/s	k_5s5p	e + Kr*(1s3) -> e + Kr(2p1) on the measured cross section out of 1s3 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
`K76r_1s3_2p1`	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_3)$	Klein–Rosseland reverse of the measured 1s3→2p1 × k_5s5p; per atom in 2p1†	1.71×10^{-8}	1.21×10^{-8}	cm^3/s	k_5s5p	e + Kr(2p1) -> e + Kr*(1s3), the reverse of K76_1s3_2p1 by the Klein-Rosseland relation on the same cross section, g(1s3)/g(2p1) = 1/1, out of the rung's level 2p1 at Tg …
`K76_1s3_2p2`	$e^- + Kr^*(1s_3) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s3→2p2 × k_5s5p	3.7×10^{-8}	8.36×10^{-8}	cm^3/s	k_5s5p	e + Kr*(1s3) -> e + Kr(2p2) on the measured cross section out of 1s3 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
`K76r_1s3_2p2`	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_3)$	Klein–Rosseland reverse of the measured 1s3→2p2 × k_5s5p; per atom in 2p2†	3.6×10^{-8}	2.83×10^{-8}	cm^3/s	k_5s5p	e + Kr(2p2) -> e + Kr*(1s3), the reverse of K76_1s3_2p2 by the Klein-Rosseland relation on the same cross section, g(1s3)/g(2p2) = 1/5, out of the rung's level 2p2 at Tg …
`K76_1s3_2p3`	$e^- + Kr^*(1s_3) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s3→2p3 × k_5s5p	1.16×10^{-7}	5.15×10^{-7}	cm^3/s	k_5s5p	e + Kr*(1s3) -> e + Kr(2p3) on the measured cross section out of 1s3 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
`K76r_1s3_2p3`	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_3)$	Klein–Rosseland reverse of the measured 1s3→2p3 × k_5s5p; per atom in 2p3†	1.88×10^{-7}	2.91×10^{-7}	cm^3/s	k_5s5p	e + Kr(2p3) -> e + Kr*(1s3), the reverse of K76_1s3_2p3 by the Klein-Rosseland relation on the same cross section, g(1s3)/g(2p3) = 1/3, out of the rung's level 2p3 at Tg …
`K76_1s3_2p4`	$e^- + Kr^*(1s_3) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s3→2p4 × k_5s5p	1.31×10^{-7}	5.77×10^{-7}	cm^3/s	k_5s5p	e + Kr*(1s3) -> e + Kr(2p4) on the measured cross section out of 1s3 (Jung, Stone, Boffard, Anderson & Lin, PRA 73, 022722 (2006) Tables II-IV). Band k_5s5p is a scale …
`K76r_1s3_2p4`	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_3)$	Klein–Rosseland reverse of the measured 1s3→2p4 × k_5s5p; per atom in 2p4†	2.03×10^{-7}	3.21×10^{-7}	cm^3/s	k_5s5p	e + Kr(2p4) -> e + Kr*(1s3), the reverse of K76_1s3_2p4 by the Klein-Rosseland relation on the same cross section, g(1s3)/g(2p4) = 1/3, out of the rung's level 2p4 at Tg …
`K76_1s2_2p1`	$e^- + Kr^*(1s_2) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s2→2p1 × k_5s5p × k_5s5p_j1	3.3×10^{-9}	6.94×10^{-9}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s2) -> e + Kr(2p1) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set …
`K76r_1s2_2p1`	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_2)$	Klein–Rosseland reverse of the measured 1s2→2p1 × k_5s5p × k_5s5p_j1; per atom in 2p1†	4.97×10^{-8}	3.57×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p1) -> e + Kr*(1s2), the reverse of K76_1s2_2p1 by the Klein-Rosseland relation on the same cross section, g(1s2)/g(2p1) = 3/1, out of the rung's level 2p1 at Tg …
`K76_1s2_2p2`	$e^- + Kr^*(1s_2) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s2→2p2 × k_5s5p × k_5s5p_j1	3.9×10^{-8}	8.41×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s2) -> e + Kr(2p2) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set …
`K76r_1s2_2p2`	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_2)$	Klein–Rosseland reverse of the measured 1s2→2p2 × k_5s5p × k_5s5p_j1; per atom in 2p2†	1.05×10^{-7}	8.32×10^{-8}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p2) -> e + Kr*(1s2), the reverse of K76_1s2_2p2 by the Klein-Rosseland relation on the same cross section, g(1s2)/g(2p2) = 3/5, out of the rung's level 2p2 at Tg …
`K76_1s2_2p3`	$e^- + Kr^*(1s_2) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s2→2p3 × k_5s5p × k_5s5p_j1	1.23×10^{-7}	5.21×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s2) -> e + Kr(2p3) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set …
`K76r_1s2_2p3`	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_2)$	Klein–Rosseland reverse of the measured 1s2→2p3 × k_5s5p × k_5s5p_j1; per atom in 2p3†	5.49×10^{-7}	8.58×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p3) -> e + Kr*(1s2), the reverse of K76_1s2_2p3 by the Klein-Rosseland relation on the same cross section, g(1s2)/g(2p3) = 3/3, out of the rung's level 2p3 at Tg …

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
K76_1s2_2p4	$e^- + Kr^*(1s_2) \rightarrow e^- + Kr^{**}$	Jung et al. measured σ, 1s2→2p4 × k_5s5p × k_5s5p_j1	1.38×10^{-7}	5.84×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr*(1s2) -> e + Kr(2p4) on the J = 1 level's own cross sections are not measured: its same-core metastable's at its own energy (1s4 the 1s5 set, 1s2 the 1s3 set …
K76r_1s2_2p4	$e^- + Kr^{**} \rightarrow e^- + Kr^*(1s_2)$	Klein–Rosseland reverse of the measured 1s2→2p4 × k_5s5p × k_5s5p_j1; per atom in 2p4†	5.92×10^{-7}	9.49×10^{-7}	cm^3/s	k_5s5p, k_5s5p_j1	e + Kr(2p4) -> e + Kr*(1s2), the reverse of K76_1s2_2p4 by the Klein-Rosseland relation on the same cross section, g(1s2)/g(2p4) = 3/3, out of the rung's level 2p4 at Tg …
KSTEP_S_1s5	$e^- + Kr^*(1s_5) \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 4.08$ eV, × kstep_s	2×10^{-9}	4.17×10^{-8}	cm^3/s	kstep_s	e + Kr*(1s5) -> Kr+ + 2e, stepwise ionization: no measurement for Kr or Ar (the 2026-10-16 survey). v0_5: Vriens & Smeets's ionization of an excited level, PRA 22, 940 …
KSTEP_S_1s4	$e^- + Kr^*(1s_4) \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 3.97$ eV, × kstep_s	2.37×10^{-9}	4.56×10^{-8}	cm^3/s	kstep_s	e + Kr*(1s4) -> Kr+ + 2e, stepwise ionization: no measurement for Kr or Ar (the 2026-10-16 survey). v0_5: Vriens & Smeets's ionization of an excited level, PRA 22, 940 …
KSTEP_S_1s3	$e^- + Kr^*(1s_3) \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 3.44$ eV, × kstep_s	5.24×10^{-9}	6.94×10^{-8}	cm^3/s	kstep_s	e + Kr*(1s3) -> Kr+ + 2e, stepwise ionization: no measurement for Kr or Ar (the 2026-10-16 survey). v0_5: Vriens & Smeets's ionization of an excited level, PRA 22, 940 …
KSTEP_S_1s2	$e^- + Kr^*(1s_2) \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 3.36$ eV, × kstep_s	5.94×10^{-9}	7.42×10^{-8}	cm^3/s	kstep_s	e + Kr*(1s2) -> Kr+ + 2e, stepwise ionization: no measurement for Kr or Ar (the 2026-10-16 survey). v0_5: Vriens & Smeets's ionization of an excited level, PRA 22, 940 …
KSTEP_P_2p10	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 2.7$ eV, × kstep_p; per atom in 2p10†	1.71×10^{-8}	1.33×10^{-7}	cm^3/s	kstep_p	e + Kr(2p10) -> Kr+ + 2e out of the rung's level 2p10 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section …
KSTEP_P_2p9	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 2.56$ eV, × kstep_p; per atom in 2p9†	2.16×10^{-8}	1.53×10^{-7}	cm^3/s	kstep_p	e + Kr(2p9) -> Kr+ + 2e out of the rung's level 2p9 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in …
KSTEP_P_2p8	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 2.55$ eV, × kstep_p; per atom in 2p8†	2.17×10^{-8}	1.53×10^{-7}	cm^3/s	kstep_p	e + Kr(2p8) -> Kr+ + 2e out of the rung's level 2p8 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in …
KSTEP_P_2p7	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 2.47$ eV, × kstep_p; per atom in 2p7†	2.49×10^{-8}	1.66×10^{-7}	cm^3/s	kstep_p	e + Kr(2p7) -> Kr+ + 2e out of the rung's level 2p7 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in …
KSTEP_P_2p6	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 2.45$ eV, × kstep_p; per atom in 2p6†	2.58×10^{-8}	1.69×10^{-7}	cm^3/s	kstep_p	e + Kr(2p6) -> Kr+ + 2e out of the rung's level 2p6 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in …
KSTEP_P_2p5	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 2.33$ eV, × kstep_p; per atom in 2p5†	3.19×10^{-8}	1.91×10^{-7}	cm^3/s	kstep_p	e + Kr(2p5) -> Kr+ + 2e out of the rung's level 2p5 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in …
KSTEP_P_2p4	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, $E_n = 1.9$ eV, × kstep_p; per atom in 2p4†	7.09×10^{-8}	3.08×10^{-7}	cm^3/s	kstep_p	e + Kr(2p4) -> Kr+ + 2e out of the rung's level 2p4 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in …

row	reaction	rate law	k, $T_e = 1$ eV	k, $T_e = 3$ eV	unit	bands	source (excerpt)
KSTEP_P_2p3	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, E_n = 1.86 eV, $\times$ kstep_p; per atom in 2p3[†]	7.66×10^{-8}	3.22×10^{-7}	cm^3/s	kstep_p	e + Kr(2p3) -> Kr+ + 2e out of the rung's level 2p3 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in . . .
KSTEP_P_2p2	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, E_n = 1.86 eV, $\times$ kstep_p; per atom in 2p2[†]	7.71×10^{-8}	3.24×10^{-7}	cm^3/s	kstep_p	e + Kr(2p2) -> Kr+ + 2e out of the rung's level 2p2 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in . . .
KSTEP_P_2p1	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	Vriens–Smeets σ, E_n = 1.74 eV, $\times$ kstep_p; per atom in 2p1[†]	9.63×10^{-8}	3.71×10^{-7}	cm^3/s	kstep_p	e + Kr(2p1) -> Kr+ + 2e out of the rung's level 2p1 at Tg (v0_5): Vriens & Smeets's ionization of an excited level, PRA 22, 940 (1980) (excited.py: its cross section in . . .
SUPER_1s5	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr$	Klein–Rosseland reverse of Kr(1S5), g ratio 0.2	5.42×10^{-10}	4.53×10^{-10}	cm^3/s		e + Kr*(1s5) -> e + Kr, G1_1s5's reverse by the Klein-Rosseland relation on its cross section (Biagi, Magboltz 8.97, trained), g(ground)/g(1s5) = 1/5. The electron gains . . .
SUPER_1s4	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr$	Klein–Rosseland reverse of Kr(1S4), g ratio 0.333	1.34×10^{-9}	1.47×10^{-9}	cm^3/s		e + Kr*(1s4) -> e + Kr, G1_1s4's reverse by the Klein-Rosseland relation on its cross section (Biagi, Magboltz 8.97, trained), g(ground)/g(1s4) = 1/3. The electron gains . . .
SUPER_1s3	$e^- + Kr^*(1s_3) \rightarrow e^- + Kr$	Klein–Rosseland reverse of Kr(1S3), g ratio 1	4.5×10^{-10}	4.18×10^{-10}	cm^3/s		e + Kr*(1s3) -> e + Kr, G1_1s3's reverse by the Klein-Rosseland relation on its cross section (Biagi, Magboltz 8.97, trained), g(ground)/g(1s3) = 1/1. The electron gains . . .
SUPER_1s2	$e^- + Kr^*(1s_2) \rightarrow e^- + Kr$	Klein–Rosseland reverse of Kr(1S2), g ratio 0.333	8.33×10^{-10}	1.1×10^{-9}	cm^3/s		e + Kr*(1s2) -> e + Kr, G1_1s2's reverse by the Klein-Rosseland relation on its cross section (Biagi, Magboltz 8.97, trained), g(ground)/g(1s2) = 1/3. The electron gains . . .
EMIX_1s5	$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^*(1s_4)$	emix_5s $T_e^0 e^{-0.117/T_e}$	3.82×10^{-8}	4.14×10^{-8}	cm^3/s	emix_5s	e + Kr*(1s5) -> e + Kr*(1s4), A exp(-0.1172 eV/Te), A the band emix_5s: Blagoev & Pilosov, Phys. Lett. A 83, 161 (1981) (abstract): 4.3e-8 cm3/s for e + 3P2 -> 3P1 in a . . .
EMIXr_1s4	$e^- + Kr^*(1s_4) \rightarrow e^- + Kr^*(1s_5)$	reverse by detailed balance (g 1.67) of emix_5s $T_e^0 e^{-0.117/T_e}$	7.17×10^{-8}	7.17×10^{-8}	cm^3/s	emix_5s	e + Kr*(1s4) -> e + Kr*(1s5), EMIX_1s5's reverse by detailed balance, g(1s5)/g(1s4) = 5/3; the gap returns to the electrons

Table 14: LIGHT_Kr v0_5's bands (network.BANDS).

band	rate	default	range	unit	rows that read it
kr2_pen	Kr2* + Kr2* -> Kr2+ + 2Kr + e	0	0 to 5×10^{-10}	cm3/s	PEN_KR2
kr2_form	Kr*(1s5) + 2Kr -> Kr2* + Kr at 300 K	3.81×10^{-32}	1.7×10^{-32} to 4.2×10^{-32}	cm6/s	AK8, AK8r
kr2_form_1s4	Kr*(1s4) + 2Kr -> Kr2* + Kr at 300 K	9.24×10^{-32}	1.27×10^{-32} to 9.24×10^{-32}	cm6/s	AK8, AK8f, AK8r, AK8fr
kr2_form_1s3	Kr*(1s3) + 2Kr -> Kr2* + Kr at 300 K	3.81×10^{-32}	0 to 9.24×10^{-32}	cm6/s	AK8, AK8r
kr2_fast	the share of the Kr*(1s4) association that forms the fast 0u+ state	0	0 to 0.17	fraction	AK8, AK8f, AK8r, AK8fr
kr2_quench	Kr2* + Kr -> 3Kr	0	0 to 8.6×10^{-15}	cm3/s	QKR2
kr2_d0	the excimer's well for its redissociation (thermo.py)	0.709	0.62 to 0.709	eV	AK8r

band	rate	default	range	unit	rows that read it
dr_tg	the gas-temperature factor on Kr2+ + e (R11p)	0	0 to 1	fraction	`R11p`
res_escape	the escaping share of the resonance radiation, a rate per Kr*(1s4) and Kr*(1s2)	0	0 to 1.7×10^{5}	1/s	`RES`
kps_5s	the share of Kr(5p) + Kr whose products go over the 5s levels by statistical weight, the rest to Kr*(1s5)	1	0 to 1	fraction	`KPS, KPSr`
kr_pen	Kr* + Kr* -> Kr+ + Kr + e	5×10^{-10}	1.5×10^{-10} to 5×10^{-10}	cm3/s	`G2`
kps_kr	Kr(5p) + Kr -> Kr(5s) + Kr	1.7×10^{-10}	3×10^{-11} to 1.7×10^{-10}	cm3/s	`KPS, KPSr`
k_5s5p	e + Kr(5s) -> e + Kr(5p), a scale on the measured cross sections out of every 5s level (jung2006.py)	1	0.6 to 1.4	factor	`K76, K76r`
k_5s5p_j1	a scale on the J = 1 levels' 5s -> 5p cross sections, their same-core metastable's sets (jung2006.py)	1	0 to 1	factor	`K76, K76r`
kstep_s	e + Kr(5s) -> Kr+ + 2e, a scale on Vriens & Smeets's calculation at each level's ionization energy (excited.py)	1	0 to 2	factor	`KSTEP_S`
kstep_p	e + Kr(5p) -> Kr+ + 2e, a scale on Vriens & Smeets's calculation at each rung level's ionization energy	1	0 to 2	factor	`KSTEP_P`
ei_kr2	e + Kr2* -> Kr2+ + 2e, a scale on Vriens & Smeets's form at the excimer's ionization energy	1	0 to 2	factor	`EI_KR2, EI_KR2F`
eq_kr2	e + Kr2* -> e + 2Kr	0	0 to 10^{-7}	cm3/s	`EQ_KR2`
em_kr2	e + Kr2* <-> e + Kr2*(0u+), the pre-exponential A of A exp(-dE/Te)	0	0 to 10^{-7}	cm3/s	`EM_KR2, EM_KR2r`
emix_5s	e + Kr(1s5) -> e + Kr(1s4), the pre-exponential A of A exp(-dE/Te)	4.3×10^{-8}	4.3×10^{-8} to 2.3×10^{-7}	cm3/s	`EMIX, EMIXr`
ib_scale	the electron-ion (Turnbull) share of the EM wave's absorbed power, a factor	1	0.9 to 1.1	factor	none (not a row rate)
hm_kr	the share of the excitations of the levels at or above E(Kr2+) (Biagi 8.97's twenty, 12.857 to 13.6 eV) that ionize associatively on a ground-state atom, Kr(nl) + Kr -> Kr2+ + e (Hornbeck-Molnar), where the rest cascade to the 5p rung	0	0 to 1	fraction	`C1c, HM`